

Energex

PUBLIC VERSION

Pricing Proposal

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positive energy



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Table of Contents

1	PREFACE	1
2	INTRODUCTION	2
2.1	Scope	2
2.2	Structure	2
2.3	Purpose	3
3	BUSINESS CHARACTERISTICS	6
4	PRICING FRAMEWORK	8
4.1	Pricing Principles and Objectives	8
4.2	Setting the 2012-13 tariffs	9
4.2.1	Maximum Allowable Revenue (MAR) and Revenue Cap	9
4.2.2	Modelling outputs	11
4.2.3	Demand, Energy Forecasts and Customer Numbers	13
5	ASSIGNMENT OF CUSTOMERS AND TARIFFS	15
5.1	Standard control services (SCS)	15
5.1.1	Assignment of customers and dispute resolution process	19
5.2	Alternative control services (ACS)	20
6	TARIFF CLASSES – STANDARD CONTROL SERVICES	22
6.1	Overview	22
6.2	Charging parameters	26
6.2.1	Fixed or standing charge parameters	26
6.2.2	Demand and Capacity charge parameters	26
6.2.3	Volume charge parameters	27
6.2.4	Recovery of DUOS	29
6.2.5	Recovery of DPPC	30
6.3	Tariffs	32
7	TARIFF CLASSES – ALTERNATIVE CONTROL SERVICES	34
7.1	Street lighting	34
7.1.1	Overview	34
7.1.2	Framework	35
7.1.3	Charging parameters	35
7.1.4	Tariffs	36
7.2	Fee-based and quoted services	36
7.2.1	Overview	36
7.2.2	Framework	37
7.2.3	Charging parameters	37
7.2.4	Tariffs	38



7.3	Large customer connections	39
7.3.1	Overview	39
7.3.2	Framework	39
7.3.3	Charging parameters	42
7.3.4	Tariffs	42
8	PROPOSED TARIFF VARIATIONS	43
8.1	Variations within the regulatory year 2012-13	43
8.2	Variations between regulatory years 2011-12 and 2012-13	43
9	TRANSMISSION COST RECOVERY	45
9.1	Expenses	46
9.1.1	Transmission charges paid to TNSPs	46
9.1.2	Avoided customer TUOS charges	46
9.1.3	Payments to other DNSPs	47
9.2	Revenue	47
9.2.1	Recovery of DPPC through tariffs	47
9.2.2	Payments received from other DNSPs	47
9.3	DPPC unders and overs account	47
10	TARIFF DEVELOPMENT	49
10.1	Structural changes to tariffs	49
10.1.1	kV.A tariffs for large customers	49
10.1.2	Tariffs for Residential Customers	50
10.2	Tariff Trials and EC&DM Programs	50
10.2.1	Controlled load tariffs	50
10.2.2	Reward-based tariff (RBT) trial	52
10.3	Future Tariff Changes	52
10.3.1	ToU demand for large customers	52
10.3.2	Appliance tariff	52
10.4	Proposed removal of tariffs	53
11	COMPLIANCE WITH REGULATORY REQUIREMENTS	54
11.1	DUOS unders and overs account	54
11.2	Compliance with avoidable and stand alone cost requirements	56
11.2.1	Standard control services	56
11.2.2	Alternative control services	59
11.3	Long run marginal cost	60
11.3.1	Standard control services	60
11.3.2	Alternative control services	61
11.4	Transaction costs	61
11.5	Response to price signals	62
11.5.1	Standard control services	62
11.5.2	Alternative control services	63
11.6	Tariff adjustment to address revenue shortfalls	63
11.7	Compliance with side constraints	63



11.8	Summary of compliance with the <i>Rules</i> and distribution determination requirements	66
12	CUSTOMER IMPACTS	71
12.1	2012-13 regulatory year	71
12.1.1	Standard control services	71
12.1.2	Alternative control services	71
12.2	Expected price trends (2011-12 to 2014-15)	73
13	ANNUAL APPROVAL PROCESS	74
14	GLOSSARY	75
14.1	Abbreviations	75
14.2	Definitions	77
15	CONFIDENTIAL INFORMATION	82
16	APPENDICES	83



1 Preface

The National Electricity Rules (the *Rules*) Section 6.18 require a Distribution Network Service Provider (DNSP) to submit an annual pricing proposal to the Australian Energy Regulator (AER), at least 2 months before the commencement of the second and each subsequent regulatory year of the regulatory control period.

This document has been prepared for the third year of Energex's 2010-2015 regulatory control period, to comply with the requirements of the *Rules* and additional requirements as specified by the AER in Energex's Final Determination 2010.



2 Introduction

2.1 Scope

This annual pricing proposal outlines the proposed network tariffs for standard control services (SCS) and alternative control services (ACS) for 2012-13.

This document is submitted in accordance with, and complies with, the requirements of:

- The National Electricity Law (NEL);
- The National Electricity Rules (*the Rules*), Version 49 (5 April 2012);
- Final Determination, Queensland, 2010-11 to 2014-15 (AER, May 2010);
- Australian Competition Tribunal Order (19 May 2011); and
- Framework and Approach Paper, Classification of services and control mechanisms, Energex and Ergon Energy 2010-15 (AER, August 2008).

2.2 Structure

Energex's annual pricing proposal is comprised of this proposal document and appendices and is structured as follows:

Table 2.1 Structure of document

Chapter	Title	Purpose
2	Introduction	Chapter 2 outlines the scope, structure and purpose of this pricing proposal.
3	Business characteristics	Chapter 3 provides an overview of the Energex business.
4	Pricing framework	Chapter 4 outlines the pricing principles and objectives which Energex has followed in setting tariffs. Chapter 4 also provides details of the modelling inputs and outputs used to develop the network tariffs to recover the regulated revenue.
5	Assignment of customers and tariffs	Chapter 5 outlines how customers and tariffs are assigned to tariff classes based on the <i>Rules</i> and pricing principles.
6	Tariff classes – standard control services (SCS)	Chapter 6 provides details of each tariff included under SCS, including a description of each tariff class and the charging parameters which are related to each tariff.



Chapter	Title	Purpose
7	Tariff classes – alternative control services (ACS)	Chapter 7 provides details of each tariff included under ACS, including a description of street lighting, fee-based and quoted services (including large customer connections) and the charging parameters which are related to each.
8	Proposed tariff variations	Chapter 8 discusses the proposed variations to tariffs during the 2012-13 regulatory year and the aspects of tariffs which have changed from 2011-12 to 2012-13.
9	Transmission cost recovery	Chapter 9 outlines how adjustments to charges for Designated Pricing Proposal Charges (DPPC) (previously transmission use of system (TUOS)) resulting from over or under recovery are calculated.
10	Tariff development	Chapter 10 outlines tariff development in the medium term, including any additional tariffs, structural changes or removal of tariffs proposed over the 2010-15 regulatory control period.
11	Compliance with regulatory requirements	Chapter 11 provides details of how this pricing proposal and the Energex methodology comply with the requirements under the <i>Rules</i> and also under the requirements of the AER Final Determination.
12	Customer impacts	Chapter 12 provides a discussion on the customer impacts as a result of the proposed tariffs for the 2012-13 regulatory year.
13	Annual approval process	Chapter 13 provides the relevant dates and details for the annual reporting and approvals.
14	Glossary	Chapter 14 defines the key terms and abbreviations used in this document.
15	Confidential Information	Chapter 15 details which parts of this document are confidential and provides reasons in support of a confidentiality claim.

2.3 Purpose

The purpose of this document is to describe the methodology and principles which Energex has followed in proposing its tariffs to recover the revenue set by the AER for the third year of the 2010-15 regulatory control period and to list the proposed prices for ACS. This is Energex's annual pricing proposal addressing the third year of the regulatory control period.



Under the requirements of the *Rules*, the following applies:

Clause 6.18.2 of the *Rules* states that:

- (a) A *Distribution Network Service Provider* must:
 - (1) submit to the AER, as soon as practicable, and in any case within 15 business days, after publication of the distribution determination, a *pricing proposal* (the "initial *pricing proposal*") for the first *regulatory year* of the *regulatory control period*; and
 - (2) submit to the AER, at least 2 months before the commencement of the second and each subsequent *regulatory year* of the *regulatory control period*, a further *pricing proposal* (an "annual *pricing proposal*") for the relevant *regulatory year*.
- (b) A *pricing proposal* must:
 - (1) set out the *tariff classes* that are to apply for the relevant *regulatory year*; and
 - (2) set out the proposed tariffs for each *tariff class*; and
 - (3) set out, for each proposed tariff, the *charging parameters* and the elements of service to which each *charging parameter* relates; and
 - (4) set out, for each *tariff class* related to *standard control services*, the expected weighted average revenue for the relevant *regulatory year* and also for the current *regulatory year*; and
 - (5) set out the nature of any variation or adjustment to the tariff that could occur during the course of the *regulatory year* and the basis on which it could occur; and
 - (6) set out how designated pricing proposal charges are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those charges in the previous *regulatory year*; and
 - (6A) set out how jurisdictional scheme amounts for each approved jurisdictional scheme are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those amounts; and
 - (6B) describe how each approved jurisdictional scheme that has been amended since the last jurisdictional scheme approval date meets the jurisdictional scheme eligibility criteria; and
 - (7) demonstrate compliance with the *Rules* and any applicable distribution determination; and
 - (8) describe the nature and extent of change from the previous *regulatory year* and demonstrate that the changes comply with the *Rules* and any applicable distribution determination.



In addition, as part of the Final Determination, the AER requested the following items to be included in Energex's annual pricing proposal:

- in accordance with clause 6.12.1(19) of the *Rules*, a record of the amount of revenues recovered from designated pricing proposal charges (DPPC)¹ and associated payments in accordance with Appendix E of the Final Determination;
- for street lighting, the proposed price for each service in the regulatory year and the revenues collected from the provision of each service in the preceding regulatory year;
- for fee-based and quoted services, the volume of services provided and revenues collected from the provision of each service in the preceding regulatory year, in addition to the proposed price for each service in the regulatory year;
- information on a distribution use of system (DUOS) unders and overs account under clause 6.18.2(b)(7) of the *Rules*; and
- information on a DPPC unders and overs account under clause 6.18.2(b)(7) of the *Rules*.

¹ Designated Pricing Proposal Charges (DPPC) previously referred to as Transmission Use Of System (TUOS)



3 Business characteristics

Energex is a Brisbane-based company with over 3900 employees and assets worth over \$9.8 billion². Energex's electricity distribution network spans approximately 25,000 square kilometres throughout South East Queensland (SEQ), stretching from Gympie in the north to Gatton in the west and Coolangatta on the New South Wales (NSW) border. This supply area encompasses the high growth regions of Brisbane, the Gold Coast and the Sunshine Coast.

Figure 3.1 Energex Area of Supply



² As reported in ENERGEX's *Annual Report 2010-2011*.

Energex's core business involves the distribution of electricity to more than 1.3 million residential, industrial and commercial connections across SEQ.

Energex maintains assets which include more than 53,000 kilometres of overhead powerlines and underground cables, more than 600,000 power poles, some 46,000 distribution transformers and over 330,000 street lights³.

Under the *Electricity Act 1994* (Electricity Act) Energex is authorised as a distribution entity with a supply area covering SEQ. Energex's entire network is classified as a distribution network for the purposes of the *Rules*.

Energex takes supply of electricity from Powerlink at transmission connection points and distributes the supply via the sub transmission and distribution network to customers throughout the region. Zone substations and distribution substations convert the voltages as necessary to minimise network losses and meet customers' voltage requirements. Energex also operates some small distributed generation which supports the network during normal and contingency situations.

Energex is committed to the objectives of the NEL in relation to the management and operation of the electricity network in the long-term interests of customers with respect to price, quality, safety, reliability and security of supply.

As an electricity distribution business operating in SEQ, the current challenges for Energex are meeting demand growth, improving security and reliability of supply, and renewal and replacement of assets within a changing operating environment.

To ensure its short-term objectives and strategies are met, Energex is required to invest significant capital and operating expenditure in supply-side solutions during the 2010-15 regulatory control period. At the same time, Energex's longer term strategies will guide the business towards delivering sustainable future performance by improving the balance between supply-side management and demand-side solutions.

³ As reported in Energex's *Annual Report 2010-2011*.



4 Pricing Framework

4.1 Pricing Principles and Objectives

Energex's overall objective in setting tariffs for SCS is to ensure that the regulated revenue is recovered from customers in a manner that is consistent with the pricing principles as outlined in the *Rules*.

For ACS (subject to a price cap), the overall objective is to ensure that the price charged is cost reflective and is consistent with the pricing principles as outlined in the *Rules*.

The pricing principles listed below are applicable to SCS and ACS.

Clause 6.18.5 of the *Rules* states that:

- (a) For each *tariff class*, the revenue expected to be recovered should lie on or between:
 - (1) an upper bound representing the stand alone cost of serving the customers who belong to that class; and
 - (2) a lower bound representing the avoidable cost of not serving those customers.
- (b) A tariff, and if it consists of 2 or more *charging parameters*, each *charging parameter* for a *tariff class*:
 - (1) must take into account the long run marginal cost for the service or, in the case of a *charging parameter*, for the element of the service to which the *charging parameter* relates; and
 - (2) must be determined having regard to:
 - (i) transaction costs associated with the tariff or each *charging parameter*; and
 - (ii) whether customers of the relevant *tariff class* are able or likely to respond to price signals.
- (c) If, however, as a result of the operation of paragraph (b), the *Distribution Network Service Provider* may not recover the expected revenue, the provider must adjust its tariffs so as to ensure recovery of expected revenue with minimum distortion to efficient patterns of consumption.

In addition to the pricing principles established under the *Rules*, Energex has applied the following pricing objectives in the formulation of tariffs. These additional objectives are intended to complement the requirements under the *Rules* and provide clarity in the formulation of robust tariffs.

- **Free from cross subsidy** – for a network user, or group of users, there should be no cross subsidies between each tariff class of SCS, or between standard control and alternative control tariffs;



- **Efficient use of the network** – tariffs should incorporate appropriate signals to network users of their impact on existing and future network capacity and costs, and to encourage demand management;
- **Equity** – tariffs should be equitable for customers and should reflect the users' utilisation of the existing network and the use of specific dedicated assets;
- **Price stability** – tariffs should not widely fluctuate over time to permit customers to make informed investment decisions;
- **Cost-reflectivity** – as far as possible, tariffs should reflect the actual cost of service provision to customers;
- **Simplicity** – tariffs should be simple and straightforward to apply, based on a well-defined and clearly explained methodology and be readily understood by customers; and
- **Designated Pricing Proposal charges (DPPC)** – where practical, network tariffs should preserve the economic signals present in the structure of DPPC.

4.2 Setting the 2012-13 tariffs

This section provides a high level overview of how the regulated revenue is recovered through tariffs for SCS. The tariffs which are set as a result of this process are provided in Section 6.

The 2012-13 tariffs for ACS are subject to a price cap form of control and are discussed in Section 7.

4.2.1 Maximum Allowable Revenue (MAR) and Revenue Cap

In May 2011, the Australian Competition Tribunal made a revised determination based on its decision that the value of gamma is 0.25. The annual revenue requirement is set out in the Tribunal's Order dated 19 May 2011.

On 12 December 2011, Energex received approval for a positive feed in tariff (FiT) direct payment pass through to be incorporated into distribution charges in 2012-13.

On 11 February 2012, Energex received a direction notice from the Queensland Government under section 115 of the *Government Owned Corporations Act 1993* (the GOC Act). The direction required Energex to exclude revenue related to the expected reduction in its capital spending program, arising from implementing the findings of the ENCAP review, from its annual network pricing proposals for the remainder of the current regulatory period.

On 4 April 2012, in response to Energex's request, the AER outlined an approach for accounting for any forgone revenues, including STPIS and ensuring that any forgone revenues are not subsequently recovered in future years. As per Energex's Statement of Corporate Intent and Corporate Plan, Energex will not be taking up the 2010-11 STPIS reward.



Table 4.1 details the revenue cap calculation for 2012-13. The revenue cap is based on a building block approach, which includes each of the regulated cost components, namely:

- regulatory depreciation;
- return on capital;
- operating expenditure; and
- tax allowance.

Table 4.1 2012-13 Revenue Cap Calculations

Component	Amount (\$m)	Comments/Reference
Annual Revenue Requirement – 2011-12	1,302.3	As approved by the AER in 2011-12 Pricing Proposal.
CPI (%) (March 2011 – March 2012)	1.58%	CPI factors as published on the ABS website.
X-Factor	-11.04%	
Impact on Revenue	166.6	
Annual Revenue Requirement – 2012-13	1,468.9	As per section 4.5.1 in Final Determination, Queensland, 2010-11 to 2014-15 (AER, May 2010).
Other Adjustments:		
Capital Contributions	0.8	Adjustment for under recovery in 2010-11
Service Target Performance Incentive Scheme (STPIS) Factor	29.4	Adjustment consistent with STPIS guidelines for performance in 2010-11
Pass-through	17.1	Approved pass-through for solar PV based on under recovery in 2010-11
MAR	1,516.2	Annual Revenue Requirement plus Adjustments
Adjustment as detailed in Energex's 2012-13 Statement of Corporate Intent (SCI)	-29.4	Removal of adjustment for 2010-11 STPIS
	-16.1	Adjustment to CAPEX due to ENCAP review
Adjusted MAR	1,470.7	Adjusted MAR
Unders/Overs Adjustments:		
DUOS	20.5	Adjustment for DUOS under recovery. Refer Table 11.1
Revenue Cap	1,491.2	MAR plus DUOS under/over adjustment

Note: individual components may not sum to the total due to rounding



4.2.2 Modelling outputs

As outlined in Table 4.1, the total Revenue Cap for 2012-13 is \$1,491.2 million. The allowances for each of the regulatory cost components are adjusted to account for the smoothed revenue and the annual adjustments as per Table 4.2 below.

Table 4.2 Redistribution of smoothed Revenue Cap to cost components

Component	Sub Total (\$m)	Total (\$m)
Regulatory Depreciation		
Original building block revenue requirement	98.1	104.8
Smoothing/CPI Allowance	1.6	
Tax Allowance	6.5	
Adjustment as detailed in Energex's 2012-13 Statement of Corporate Intent (SCI)	-1.5	
Return on Capital		
Original building block revenue requirement	987.3	1,000.6
Smoothing/CPI Allowance	0.8	
Tax Allowance	65.8	
Solar PV Pass-through	17.1	
Adjustment as detailed in Energex's 2012-13 Statement of Corporate Intent (SCI)	-14.7	
Capital Contributions	-71.5	
Revenue from Shared Assets	-5.5	
Unders/Overs Adjustments:		
DUOS	20.5	
Capital Contributions	0.8	
STPIS	-	
Operating Expenditure		
Original building block revenue requirement	354.7	385.8
Smoothing/CPI Allowance	7.4	
Tax Allowance	23.6	
Adjustment as detailed in Energex's 2012-13 Statement of Corporate Intent (SCI)	-	
Revenue Cap	1,491.2	1,491.2

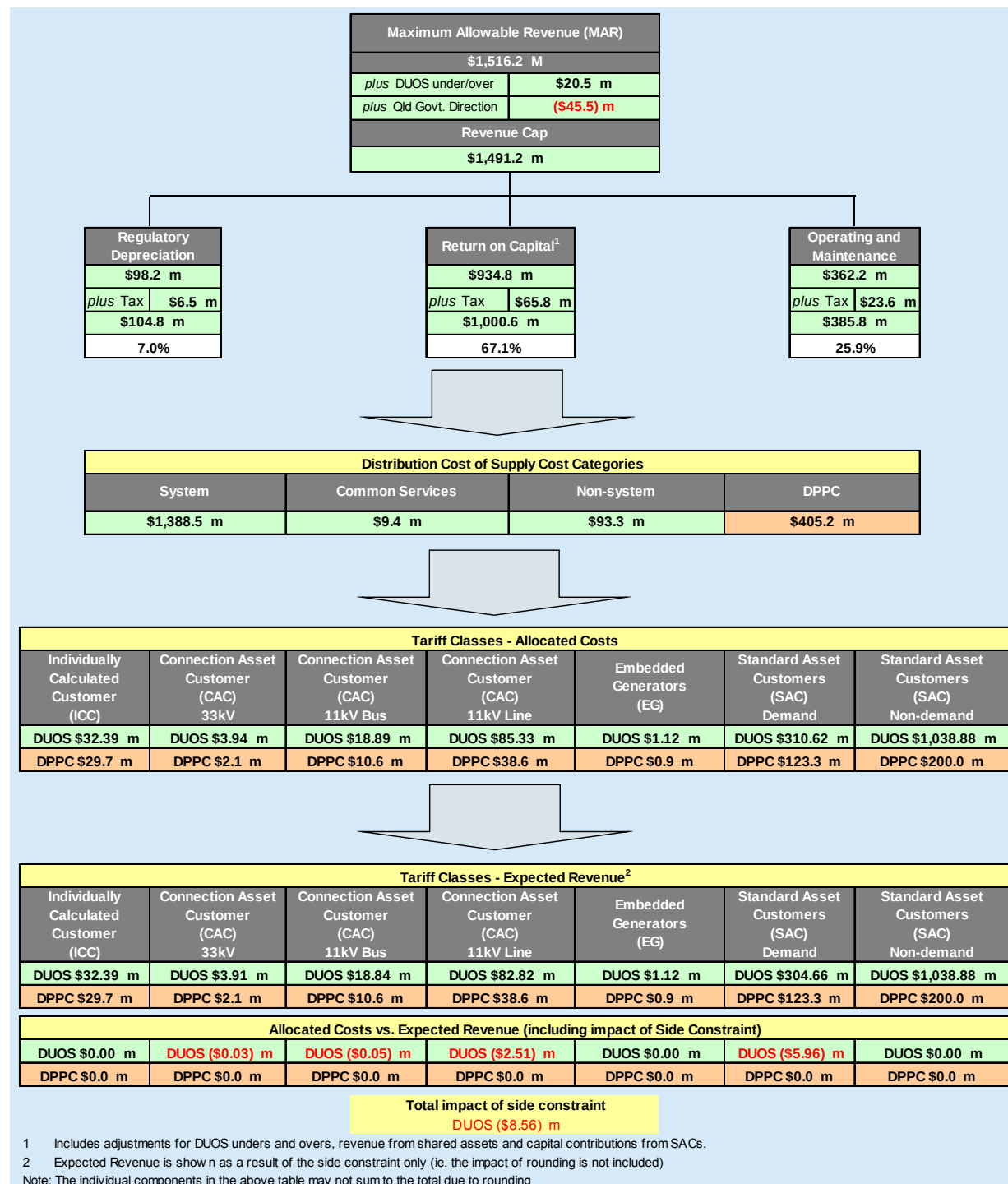
Note: individual components may not sum to the total due to rounding

The first stage of the tariff determination process is to allocate or assign these network costs to the tariff classes in the most efficient and cost reflective way. Energex allocates costs to



tariff classes using a Distribution Cost of Supply (DCOS) model. This modelling process is explained in Appendix 3. The allocation of costs to recover the revenue cap is illustrated in Figure 4.1. Due to side constraints, the expected revenue for 2012-13 is \$1,482.6 million.

Figure 4.1 2012-13 Tariff Development



4.2.3 Demand, Energy Forecasts and Customer Numbers

As part of the *Regulatory Information Notice* (RIN) for the AER Final Determination (May 2010), Energex provided the AER details of the demand and energy forecasts and expected customer numbers over the regulatory control period. A set of forecasts were approved in the AER Final Determination (May 2010). The approved forecast figures from the Final Determination have now been revised, as shown in Table 4.3 below.

The revised energy forecast follows an unanticipated decline in energy purchases for the current (2011-12) financial year, primarily as a result of depressed energy sales over the financial year following the mild winter, and more recently, a mild summer season associated with the La Nina climate phenomenon.

Energy sales for 2011-12 are forecast to reach 21,715 GW.h. This is very similar to the 2010-11 sales. Energex's initial assessment has determined that the decrease is primarily related to the very wet and mild summer season and a delayed economic resurgence. However, there are likely to be a number of other contributing factors including the impact of solar PV and possibly customer reaction to recent electricity price increases.

Using this figure as the basis for the 2012-13 energy sales forecast, gives a likely energy sales value of 22,096 GW.h (a 1.75% increase on revised 2011-12 figures). This reflects a more optimistic outlook for consumption growth.

The estimated customer numbers for large customers (ICC and CAC) are based on current forecast information. The total customer numbers estimated for small customers (SAC) in 2012-13 are based on actual customer numbers at 31 December 2011 obtained from network billing systems, with a small allowance for population growth.

Forecasts of demand, energy and customer numbers are used to allocate different costs to tariff classes. This is described further in the tariff cost allocation process (Appendix 3).



Table 4.3 2012-13 demand, customer and energy forecasts by tariff class

Tariff Class	Maximum Demand (MW)	Forecast Customer Numbers	Forecast Energy Consumption (GW.h)
2012-13 Original Forecast (Approved in Final Determination)	5,328	1,417,664	24,042
2012-13 Revised Forecast	5,359 ⁴	1,353,560	22,096
<i>Split of Revised Forecast into tariff classes:</i>			
ICC	280	33	1,794
CAC – 33kV	40	10	119
CAC – 11kV Bus	75	52	867
CAC – 11kV Line	415	394	2,879
EG	15	18	56
SAC Demand	1,147	8,813	5,287
SAC Non-Demand	3,387	1,344,240	11,094
Total	5,359	1,353,560	22,096

⁴ As per latest published *Energex Network Management Plan – 2011/12 to 2015/16*



5 Assignment of customers and tariffs

5.1 Standard control services (SCS)

This section outlines Energex's approach to the assignment and reassignment of customers to tariff classes to comply with clause 6.18.4 of the *Rules*.

Each customer for SCS is a member of at least one tariff class. Energex's customer assignment to tariff classes is determined based on assessment of one or more of the following criteria:

- a) energy consumption and/or demand; and
- b) voltage level.

To provide the appropriate economic and cost of supply signals, the tariff regime distinguishes between different tariff classes. Energex has established tariff classes based on the criteria above, and takes into account the relevant customers' capability to respond to price signals as well as their capacity to make investment and operational decisions that will affect their electricity usage profile.

In addition to the above, the following guidelines apply:

- allocation of a customer with micro-generation facilities to a tariff will be made on the same basis as other connections;
- where a new tariff is applied to a customer, it is standard practice to apply the tariff from the next billing period; and
- for new connections with no previous load history, the default tariff is based on their expected energy usage, supply voltage and meter type.

In accordance with clauses 6.18.4(a)(4) and 6.18.4(b), assignment of customers to tariffs is reviewed periodically to assess if the tariff assignment is still applicable given potential changes in annual usage. A change in connection voltage means that the connection is treated as if it was a new connection.

To limit transaction costs and ensure pricing signals are not distorted by constant changes in customer tariff assignment, customers are generally only allowed one tariff change per 12 month period.

For customers with demand levels which fluctuate year to year, Energex may apply a tolerance limit of up to 20 per cent around tariff thresholds to mitigate frequent tariff reassignment and subsequently limit customer impact.



The procedure for assigning and reassigning customers to tariff classes satisfies the requirements of clause 6.18.4. This procedure relates specifically to the application of mandated tariffs. Customers who have chosen to be assigned to a voluntary tariff will not normally be subject to this review process.

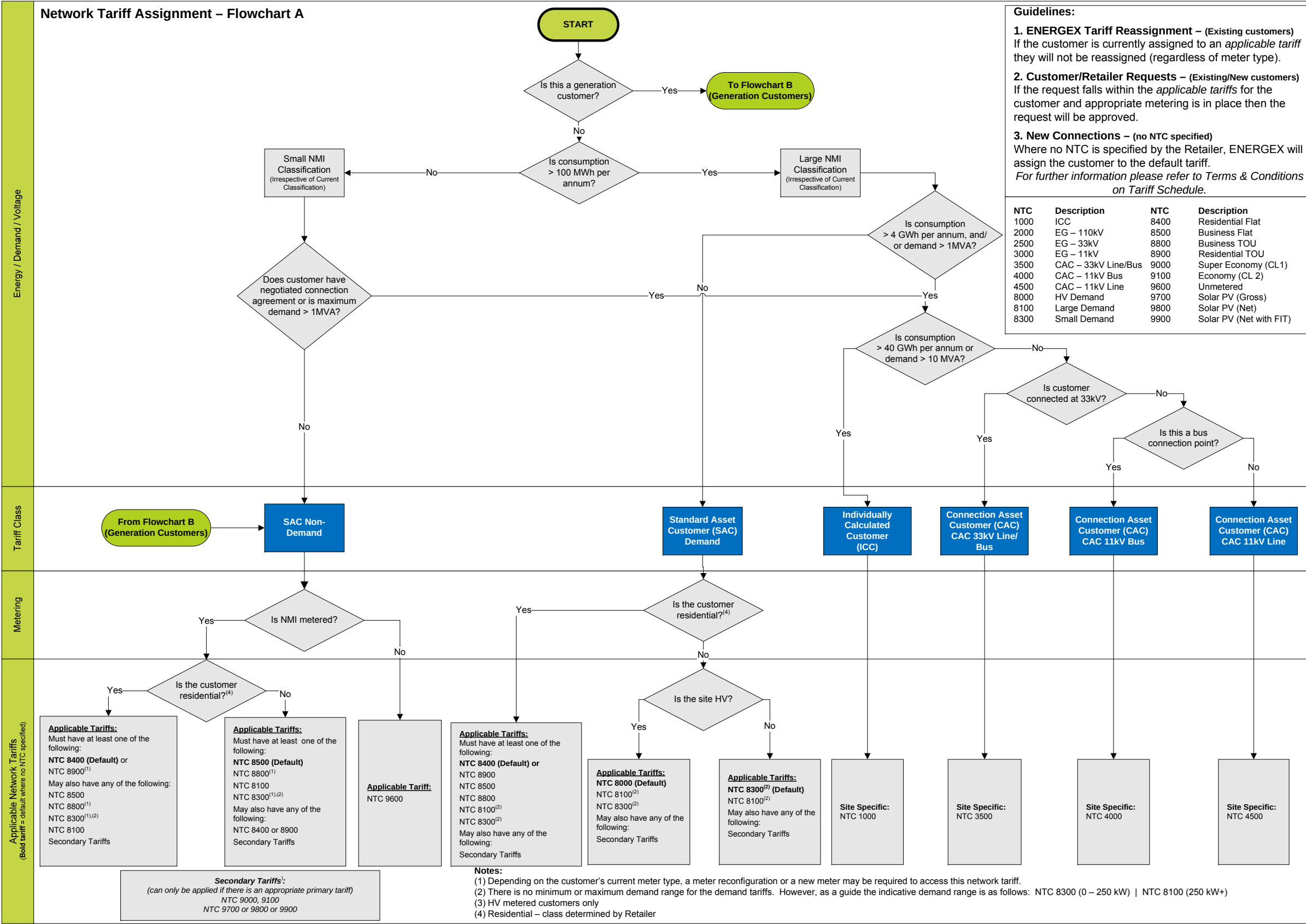
Energex believes this approach satisfies the requirements of clauses 6.18.3(d) and 6.18.4, namely:

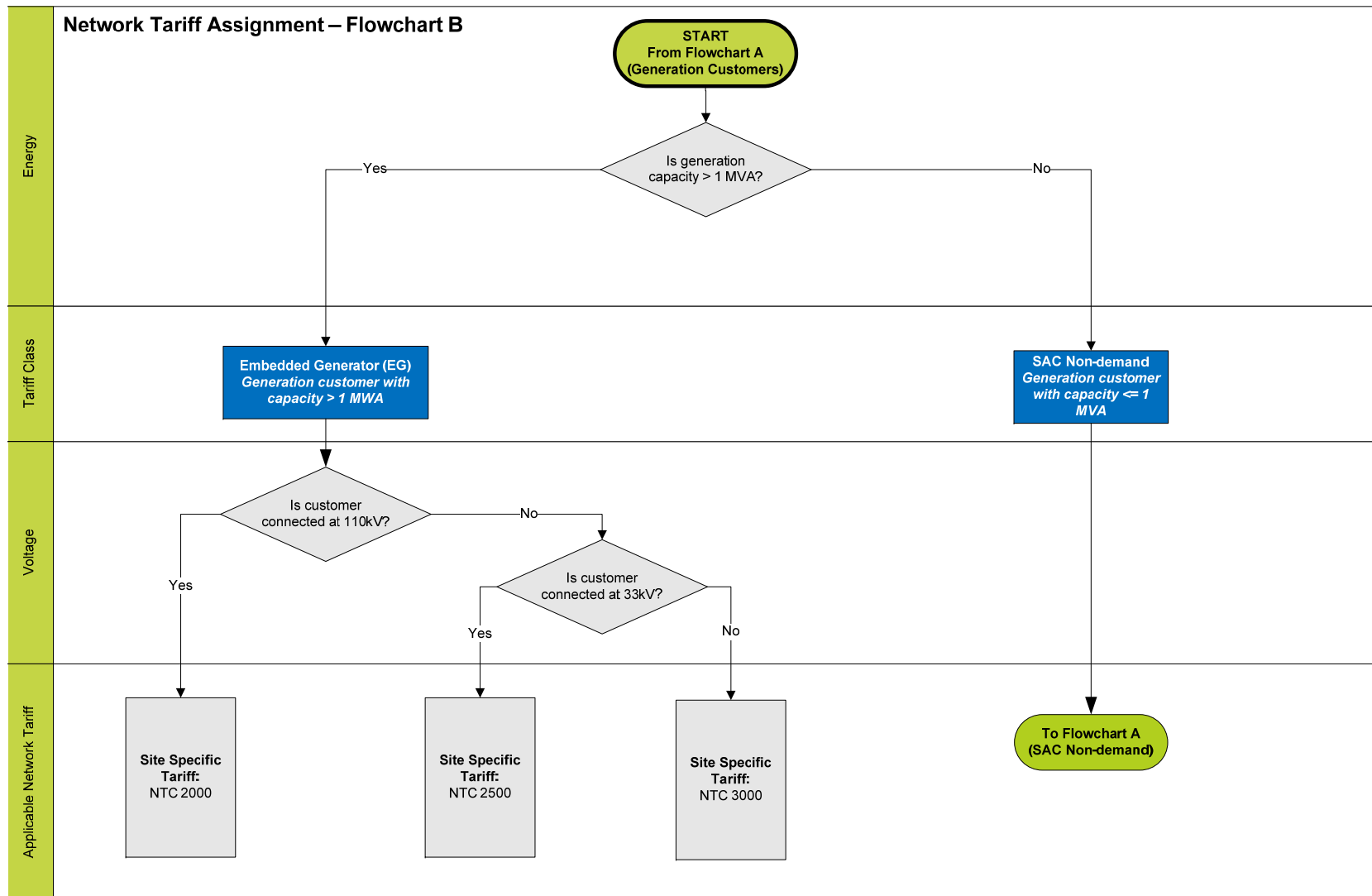
- balancing the grouping of customers on an economically efficient basis with the need to avoid unnecessary transaction costs; and
- ensuring customers with a similar connection and usage profile are treated on an equal basis and customers with micro-generation facilities are not treated less favourably.

A pictorial representation of the process for assignment to tariff class for SCS is outlined in Figure 5.1.



Figure 5.1 Assignment of customers to tariff classes – standard control services





5.1.1 Assignment of customers and dispute resolution process

In accordance with Appendix B of the Final Determination, customers (and their Retailer) will be notified of the tariff class to which they have been assigned or reassigned. In summary, the process for notifying customers is as outlined in Table 5.1.

Table 5.1 Customer notification process for tariff class changes

Input to Tariff Class Assignment Process	Notification Process
Energex driven reassignment based on a change in usage or connection	Energex identifies customers who are assigned to an incorrect tariff class. The correct tariff class is determined based on the process outlined in Section 5. The customer is notified in writing of the reclassification, and is given the opportunity to object.
Customer driven reassignment (through Energex Form 1634)	Energex receives a completed Form 1634 (QES19) from the customer/Retailer. If the request is approved: ▪ The customer is notified of the reclassification in writing. If the request is not approved: ▪ The customer is notified that the reclassification has not been approved. ▪ The customer is given the opportunity to object to the decision.
New connection	Energex receives notification of a new customer connection. For CAC, EG and ICC customers: ▪ The correct tariff class is determined based on the process outlined in Section 5. ▪ The customer is notified of the tariff classification as part of the Connection Agreement, and is given the opportunity to object. For SAC Demand and SAC Non-Demand customers: ▪ Where a tariff class is nominated on the connection request, Energex will confirm if this is appropriate. ▪ If a tariff class is not nominated on the connection request, the correct tariff class is determined based on the process outlined in Section 5. ▪ Notification to the Retailer will occur electronically by way of a Change Request notice through Market Settlement and Transfer Solution (MSATS).



Where the customer contact details are known, the customer will be notified directly. If the customer details are unknown, (or if the letter is returned to Energex), the letter will be sent to their Retailer.

The notice given will include advice that the customer may request further information from Energex and that they may object to the proposed assignment or reassignment.

This notice will specifically include:

- a) either a copy of Energex's internal procedures for reviewing objections or the link to where such information is available on the Energex website;
- b) that if the objection is not resolved to the satisfaction of the customer under Energex's internal review system, then to the extent that resolution of such disputes is within the jurisdiction of a state based energy ombudsman scheme the customer is entitled to escalate the matter to such a body; and
- c) that if the objection is not resolved to the satisfaction of the customer under Energex's internal review system, then the customer is entitled to seek resolution via the dispute resolution process available under Part 10 of the NEL.

If a customer makes an objection to Energex about the proposed assignment or reassignment, Energex will reconsider the proposed assignment or reassignment, taking into consideration clause 6.18.4(a)(1) to (3) of the *Rules*, and section 5 of the Pricing Proposal. Energex will notify the customer in writing of its decision and the reasons for that decision.

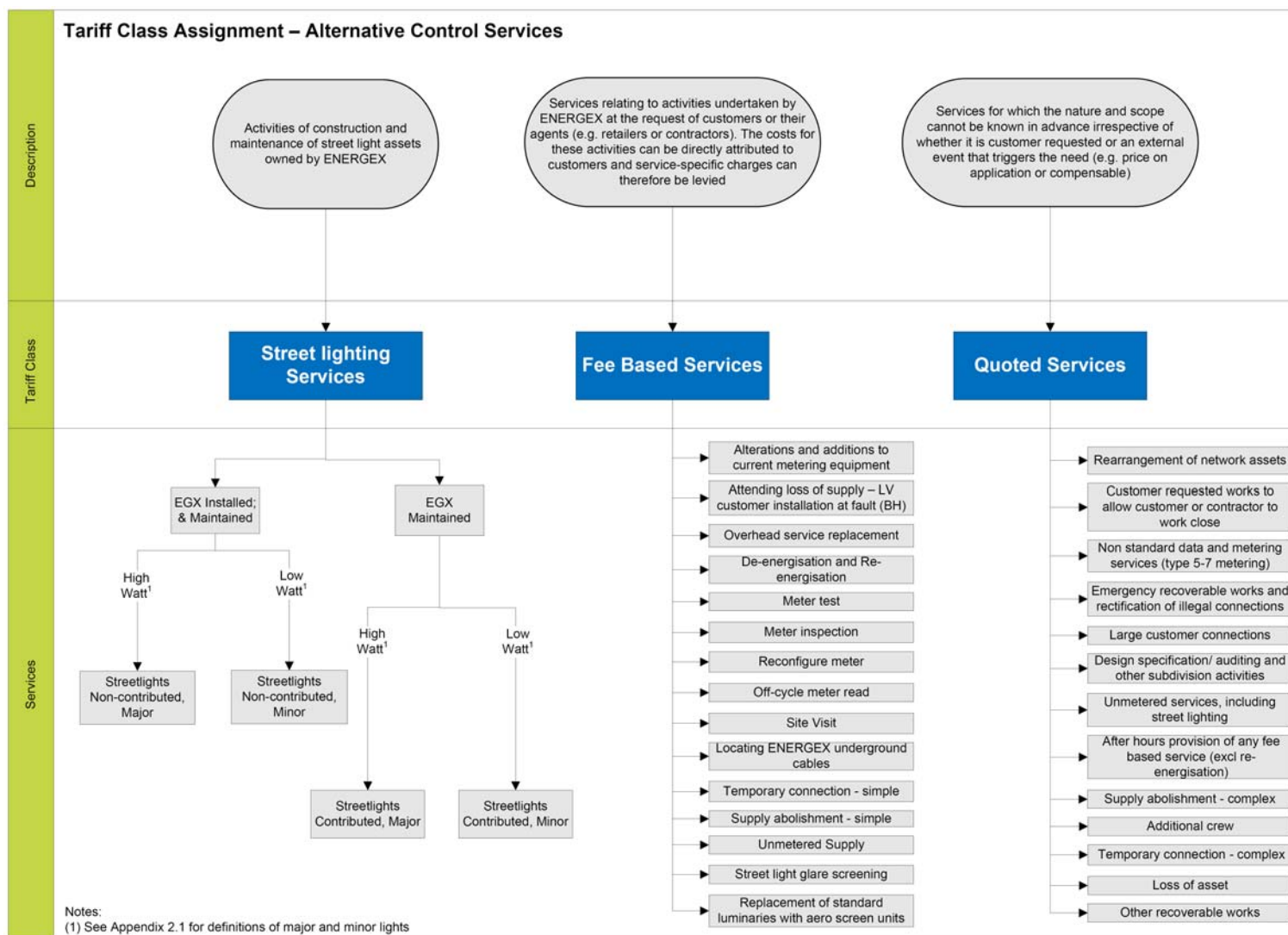
5.2 Alternative control services (ACS)

The assignment of customers to an ACS tariff is based on the type of service required. Each customer for ACS is a member of at least one tariff class.

Figure 5.2 below outlines the tariff class and service types which are included under ACS.



Figure 5.2 Assignment of customers to tariff classes – ACS



6 Tariff classes – standard control services

In the Final Framework and Approach paper⁵, the AER classified network services, connection services and metering services as SCS. This pricing proposal is in accordance with the AER's decision.

6.1 Overview

In line with clause 6.18.3(d) of the *Rules*, tariff classes have been selected in recognition of the need to group together customers on an economically efficient basis which adequately reflects customer characteristics with regard to transaction costs. The tariff classes for SCS are outlined in Table 6.1.

⁵ Final decision - framework and approach - classification of services and control mechanisms to Energex and Ergon Energy, 27 August 2008, <http://www.aer.gov.au/content/index.phtml/itemId/721654>



Table 6.1 Tariff classes for SCS

Tariff Class	Description	Tariff Details
ICC	The ICC tariff class applies to customers with electricity consumption greater than 40 GW.h per year at a single connection point; or where the customers demand is greater than or equal to 10 MVA; or where a customer's circumstances mean that the average shared network charge becomes meaningless or distorted. Tariffs are based on actual DPPC charges for the relevant transmission connection point, plus charges associated with the customer's shared distribution network utilised for the electricity supply plus connection charges based on the customer's connection assets. This provides the greatest cost reflectivity for these customers and is feasible since the number of such customers is relatively small. Where there is a network on private property and there are site specific Energex costs associated with operating, maintaining and accessing the network, these costs should be applied directly to the users of those assets when it is economically efficient to do so. The classification of customers as an ICC with electricity consumption lower than 40 GW.h per year at a single connection point ,will be at Energex's discretion, and could occur where: ▪ A customer has a dedicated supply system which is quite different and separate from the remainder of the supply network; ▪ There are only two or three customers in a supply system making average prices inappropriate; or ▪ Inequitable treatment of otherwise comparable customers arises from the application of the 40 GW.h cut-off.	The tariffs for ICCs are provided in Appendix 1.1 (Confidential). Tariffs are calculated on a site-specific basis and are not published due to confidentiality. Energex will provide site-specific tariffs directly to the customer and their Retailer.
CAC 33kV Line/Bus	The CAC tariff classes including 33kV, 11kV Line and 11kV Bus, applies to customers with electricity consumption greater than 4 GW.h (but less than 40 GW.h per year) at a single connection point; or where demand is greater than or equal to 1MVA at a single connection point; or where a customer has a dedicated supply system with significant connection assets; or where the customer has contributed to their dedicated connection assets.	Tariffs for CACs include a mix of site specific charging parameters (fixed charge) and general tariff class charging parameters (capacity, demand and volume charges). Tariffs for the site-specific charging parameter (fixed charge) are provided in Appendix 1.2 (Confidential). Energex will provide these site-specific charges directly to the customer and their Retailer. The other charging parameters for CACs are outlined in Table 6.5
CAC 11kV Line	Tariffs are based on average DPPC charges, plus average shared network charges plus site-specific connection charges based on the customer's connection assets. This provides a significant degree of cost reflectivity for this group of customers while recognising the practical difficulties of calculating individual shared network charges for each customer. Where there is a network on private property and there are site specific Energex costs associated with operating, maintaining and accessing the network, these costs should be applied directly to the users of those assets when it is economically efficient to do so.	
CAC 11kV Bus	The classification of customers as a CAC with electricity consumption lower than 4 GW.h per year at a single connection point, will be at Energex's discretion, and could occur where: ▪ A customer has a maximum demand > 1MVA; ▪ A customer has a dedicated supply system which is quite different and separate from the remainder of the supply network; or ▪ Inequitable treatment of otherwise comparable customers arises from the 4 GW.h per year cut-off.	

Tariff Class	Description	Tariff Details
Embedded Generator (EG)	The Embedded Generator (EG) tariff class applies to those generators with an installed capacity > 1 MVA in accordance with the Energy Networks Association (ENA) definition as follows: ▪ Medium: 1-5 MVA (LV or HV) or < 1 MVA (HV); and ▪ Large: > 5 MVA. Tariffs for connection and access services for Medium and Large EGs will be developed on a similar basis to site-specific customers. This is due to the nature of connections, which are typically non-standard and may require additional embedded generator protection system upgrades. In accordance with the <i>Rules</i>, all EGs will receive a charge for connection services regardless of whether they are a net importer or exporter of electricity. However, DUOS charges will not be incurred for the export of electricity generated by the user into the distribution network. EGs who are net importers of electricity will receive appropriate network charges. Micro and Small generators with an installed capacity <= 1 MVA (LV) will be connected under an applicable SAC tariff based on the consumption level and metering type of the customer. However, a small generator may be treated as a CAC if there is a supply arrangement which is different to that which would be provided for a load customer of similar size or as per CAC requirements.	Tariffs for EGs include a mix of site specific charging parameters (fixed charge) and general tariff class charging parameters (capacity, demand and volume charges). Tariffs for the site-specific charging parameter (fixed charge) are provided in Appendix 1.2 (Confidential). Energex will provide these site-specific charges directly to the customer and their Retailer. The other charging parameters for EGs are outlined in Table 6.5
SAC Demand	The SAC Demand tariff class applies to those customers with consumption less than 4 GW.h per year. Customers must have a meter installed that is capable of measuring energy consumption (kW.h) and demand (kW). This meter records total energy consumption and demand over 30 minute periods. Each tariff has been built to include a demand parameter so that the customer's actual demand is reflected in the price that they pay for their electricity. The highest demand reading for that month is used to calculate the customer's electricity bill. Tariffs with a demand element are generally more cost reflective compared to those with only a volume (energy) charge. SAC Demand tariffs include the following: • HV Demand: This tariff is available for customers connected at HV with demand up to 1000 kVA. • Large Demand: This tariff is generally applies for customers with maximum demand between 250 and 1000 kVA. • Small Demand: This is the default tariff for customers between 100 MW.h and 4 GW.h per year where no previous demand history exists. Generally the small demand tariff applies for customers with maximum demand between 0 and 275 kVA. Customers with consumption less than 100 MW.h can choose to access this tariff on a voluntary basis.	Tariffs are based on an average shared network prices and average connection price. The tariffs for SACs are outlined in Table 6.5. Capital contributions apply primarily to newly connecting Standard Asset Customers (SACs), and are sought as prepayment for a revenue shortfall in the case of an uneconomic connection. Energex's capital contributions policy is available on the Energex website.

Tariff Class	Description	Tariff Details
SAC Non-demand	The Non-demand SAC tariff class applies to customers with consumption below 100 MW.h per year. A non-demand tariff may also apply when the customer's connection point has a meter installed that is capable of measuring total energy consumption (kW.h) only. Non-Demand tariffs include the following: • Business Flat and Business Time-of-Use (ToU): These tariffs are available for customers less than 100 MW.h per annum. The ToU tariff takes into account when, as well as how much, electricity is used by each consumer. With ToU, electricity is priced at multiple levels, depending on the time of day. Tariffs are lower during off-peak hours and higher during peak hours. • Residential Flat: The residential Flat tariff is the default tariff for residential customers, regardless of their size. • Residential ToU: The residential ToU tariff is a voluntary tariff for residential customers, regardless of their size. Customers must have a ToU capable meter to access this tariff. • Solar PV: ▪ Solar PV (net metering – feed-in tariff) – this scheme applies to SACs who consume less than 100MW.h per annum. The Queensland Government sets the feed-in tariff rate (per kW.h) to be paid for the excess energy generated. Energex pays this feed-in tariff; ▪ Solar PV (net metering – no feed-in tariff) – where the electricity Retailer purchases the excess energy generated. No feed-in tariff is provided by Energex; and • Solar PV (gross metering) – where all energy generated is purchased by the electricity Retailer. No feed-in tariff is provided by Energex. • Super Economy (controlled load 1): Specified connected appliances are controlled by network equipment so that supply will be permanently available for a minimum period of 8 hours at the absolute discretion of Energex but usually between the hours of 10:00pm and 7:00am. • Economy (controlled load 2): Specified connected appliances are controlled by network equipment so that supply will be available for a minimum period of 18 hours per day during time periods set at the absolute discretion of Energex. • Unmetered: This tariff is applicable to unmetered supplies. This includes facilities such as street lighting, watchman lights, public telephones, traffic signals, and public barbecues. Energex only provides connection to the network for these services. The unmetered supply tariff therefore seeks to only recover a contribution towards the shared network (use of system charge). For street lighting and watchman lights, additional costs may be incurred for the capital and maintenance cost of the assets (this will be recovered as an ACS). For more detailed terms and conditions relating to each tariff please refer to Energex's Tariff Schedule.	Tariffs are based on an average shared network prices and average connection price. The tariffs for SACs are outlined in Table 6.5. Capital contributions apply primarily to newly connecting Standard Asset Customers (SACs), and are sought as prepayment for a revenue shortfall in the case of an uneconomic connection. Energex's capital contributions policy is available on the Energex's website.



6.2 Charging parameters

Clause 6.18.2(b)(3) of the Rules states that:

A pricing proposal must: set out, for each proposed tariff, the charging parameters and the elements of service to which each charging parameter relates.

Consistent with the *Rules*, the tariffs proposed by Energex will be comprised of a number of charging parameters, each selected and structured to provide signals to customers about the efficient use of the network and their impact on future network capacity and costs.

The tariff structure and the proportioning of charge parameters have been developed to achieve the pricing principles and objectives listed previously. In developing the tariffs, Energex has sought to select tariff parameters which signal the impact that the end users will have on the network whilst managing the demand and volume variance risk, minimising boundary issues between and within tariff classes and avoiding any signals that may result in perverse outcomes.

Energex provides for the following charging parameters in the tariffs:

6.2.1 Fixed or standing charge parameters

Fixed charges reflect the incremental costs that arise from the connection and management of the customer. This sends a clear signal about the economic value of the dedicated connection and provides a stable price and income regardless of the usage of the network. Network usage signals are provided by other charging parameters.

6.2.2 Demand and Capacity charge parameters

Demand charges are reflective of augmentation costs associated with customer demand activity. There are two demand charge parameter types:

- a monthly maximum demand charge is levied on the basis that network users who place greater pressure on the system should incur higher charges. Network expansion becomes necessary where there is a likelihood of demand exceeding available capacity; and
- a capacity charge *is* similar to a monthly maximum demand charge, but more effectively assigns an adequate share of costs associated with system augmentations to network users. The capacity charge reflects the amount of network which is set aside for the customer which could be used by the customer at any time.

Demand charges send price signals to customers that they can reduce their network charges by reducing the variability of their consumption, as a lower peak demand reduces the pressure that they place on the network. Demand charges account for augmentation costs at the customer connection level and all associated upstream augmentation costs. For example, if a network user is connected at the low voltage level, the ability of the system to meet an increase in that user's demand is dependent upon the available capacity of assets



further up the distribution network – low voltage, distribution, zone substations and sub-transmission.

Potentially, customers connected at a lower voltage distribution level, will require a greater capital expenditure for supply (due to a greater proportion of upstream costs), and the greater will be the demand charge levied.

6.2.3 Volume charge parameters

Volume charge parameters have been adopted to recover those costs that are related to the size of the customer but not specifically the demand of that customer on the network. These costs include overheads and common service costs. Volume charges are also used to recover any residual allocated costs after the capacity and demand charges have been calculated. For non-demand customers the volume charge parameter also recovers those costs that would have been recovered from a demand based charge. A volume charge parameter can be structured either as flat anytime energy charge or a Time of Use (ToU) energy charge.

6.2.3.1 Time of Use energy charge

The objective of a ToU energy charge is to reduce the demand on the network during peak times by encouraging customers to switch non-essential electricity use to off-peak or shoulder periods. This will reduce the infrastructure expenditure required to meet increasing peak demand, ensuring resources are used more efficiently which will ultimately benefit all customers through reduced network costs over the long term.

The ToU tariff provides a price signal by offering a lower tariff during off-peak periods and higher tariffs during peak periods.

ToU tariffs were introduced for small and medium non-demand business customers from 1 July 2007 and for large business customers from 1 July 2008.

A new, voluntary ToU tariff is being introduced this year for residential customers. ToU tariffs have not been offered to residential customers in the past due to metering limitations. However, a number of residential customers now have meters capable of ToU pricing in Energex's area. All new meters installed since 2007 are capable (once programmed) to record energy consumption at different periods of the day.

The voluntary ToU for residential customers will provide more cost reflective pricing for residential customers and bring Energex in line with pricing structures in other DNSPs. Larger residential consumers will also be encouraged to switch from the default flat rate to the voluntary ToU tariff which will provide incentive for users to decrease consumption during periods of peak demand.

The current times for the business ToU energy tariffs and proposed times for the residential ToU energy tariffs are as follows:



Table 6.2 Times for Energex ToU tariffs

Tariff	Tariff Element	Weekdays	Weekends
Residential ToU	Off-Peak	10pm – 7am	10pm – 7am
	Shoulder	7am – 4pm, 8pm – 10pm	7am – 10pm
	Peak	4pm – 8pm	No Peak
Business ToU (Non-Demand tariff)	Off-Peak	9pm – 7am	Anytime
	Peak	7am – 9pm	No Peak
Business ToU (ICC, CAC, EG)	Off-Peak	11pm – 7am	Anytime
	Peak	7am – 11pm	No Peak



6.2.4 Recovery of DUOS

Network tariffs and charging parameters are designed to recover the approved revenue which is consistent with the calculation of the MAR as set out in Section 4. The network charging parameters adopted by Energex for the recovery of DUOS for SCS tariffs are detailed in Table 6.3.

Table 6.3 Tariff charging parameters for DUOS charges

Tariff Class	Tariff	Tariff Charging Parameter					
		Fixed or Service Availability Charge (\$/day)	Capacity Charge ^{1,2} (\$/kW/month)	Monthly Maximum Demand Charge ¹ (\$/kW/month)	Power Factor Adjustment (PFA) for Capacity & Demand ³	Volume Charge Flat (¢/kW.h)	Volume Charge ToU (¢/kW.h)
ICC	ICC	✓	✓	✓	✓		✓
CAC – 33kV Line/Bus	CAC – 33kV Line/Bus						
CAC – 11kV Line	CAC – 11kV Line	✓	✓	✓	✓		✓
CAC – 11kV Bus	CAC – 11kV Bus						
EG	EG	✓	✓	✓ ⁵			✓ ⁴
SAC Demand	HV Demand						
	Small Demand	✓		✓		✓	
	Large Demand						
SAC Non-demand	Business Flat	✓				✓	
	Business ToU	✓					✓
	Residential Flat	✓				✓	
	Residential ToU	✓					✓
	Solar PV	Not applicable (Solar PV is subject to a feed in tariff only)					
	Super Economy (controlled load 1)					✓	
	Economy (controlled load 2)					✓	
	Unmetered					✓	

1. The application of capacity or demand-based charges (either authorised or actual) is limited by the type of metering installed. Demand charges are not appropriate for those customers with metering equipment only capable of measuring and recording energy consumption.
2. In general, the capacity charge is levied on the basis of either contracted demand or the maximum demand in the previous calendar year.
3. From 1 July 2010, charges for ICCs and CACs are adjusted up or down by a relative amount based on the difference between the customer's measured power factor and the power factor neutral point. The adjustment will be applied to the demand and capacity charge elements.
4. These rates apply to EGs only when they are importing electricity to their site (i.e. as a consumer).



6.2.5 Recovery of DPPC

Most electricity is delivered from Generators to Energex's network via Powerlink's transmission network. Energex pays DPPC to Powerlink on behalf of its customers and recovers these costs through the network tariffs.

Energex's transmission cost recovery tariffs are based on a forecast of DPPC for each year, adjusted for under or over recoveries. The DPPC from Powerlink comprise both fixed and variable charges. Where administratively efficient and locational signals are material, the forecast DPPC will be passed on to customers in the same form of price structure as received from Powerlink. A forecast of the DPPC is provided by Powerlink in March of each year. This allows Energex to develop tariff components for recovery of the anticipated costs. The network charging parameters adopted by Energex for DPPC services are detailed in Table 6.4.

In July 2012, a new pricing methodology for Powerlink will be introduced which significantly alters the structure of locational network charges. This change has been detailed in Table 6.4 below.



Table 6.4 Tariff charging parameters for DPPC

Tariff Class	Tariff	Tariff Charging Parameter			
		Fixed or Service Availability Charge (\$/day)	Monthly Maximum Demand Charge ¹ (\$/kW/month)	Volume Charge Flat (¢/kW.h)	Volume Charge ToU (¢/kW.h)
ICC	ICC	✓	✓ ²	✓ ³	
CAC – 33kV Line/Bus	CAC – 33kV Line/Bus	✓	✓	✓	
CAC – 11kV Line	CAC – 11kV Line				
CAC – 11kV Bus	CAC – 11kV Bus				
EG	EG	✓	✓ ⁴	✓ ⁴	
SAC Demand	HV Demand	✓	✓	✓	
	Small Demand				
	Large Demand				
SAC Non-demand	Business Flat	✓		✓	
	Business ToU	✓			✓
	Residential Flat	✓		✓	
	Residential ToU	✓			✓
	Solar PV				
	Super Economy (controlled load 1)			✓	
	Economy (controlled load 2)			✓	
	Unmetered			✓	

1. The application of demand based charges (authorised), is limited by the type of metering installed. Demand charges are not appropriate for those customers with metering equipment only capable of measuring and recording delivered electricity volume.
2. Monthly maximum demand charge for ICC's is the locational charge as published by Powerlink and consists of the nominated demand plus average demand multiplied by rate.
3. Volume charge for ICC's is a combination of General and Common Charge as published by Powerlink.
4. These rates apply to EGs only when they are importing electricity to their site (that is a load customer)



6.3 Tariffs

Clause 6.18.2(b)(2) of the *Rules* states that:

A pricing proposal must: set out the proposed tariffs for each tariff class.

The proposed tariffs for SCS are provided in Table 6.5.

Site specific tariffs for ICC, CAC and EG customers are provided in Appendix 1. The tariffs for SCS are set at the beginning of the regulatory year. However, within a regulatory year there may sometimes be a requirement to include a new site specific charge. If a new charge is required, the charge will be calculated in accordance with the approved pricing proposal.



Table 6.5 Proposed tariffs for DUOS and DPPC (TUOS) – standard control services

Tariff Class	Tariff Description	Network Tariff Code	DUOS Charges ^a							DPPC Charges ^a					
			Fixed (\$/day)	Capacity (\$/kW/month)	Demand (\$/kW/month)	Volume Flat (c/kW.h)	Volume Off Peak (c/kW.h)	Volume Shoulder (c/kW.h)	Volume Peak (c/kW.h)	Fixed (\$/day)	Demand (\$/kW/month)	Volume Flat (c/kW.h)	Volume Off Peak (c/kW.h)	Volume Shoulder (c/kW.h)	Volume Peak (c/kW.h)
ICC	ICC	1000	Tariffs for ICC customers are confidential and are provided in Appendix 1.1.												
CAC 33kV Line/Bus	CAC 33kV Line/Bus	3500	Site Specific (see Appendix 1.2)	1.063	2.596		0.058		0.073	Site Specific (see Appendix 1.2)	1.112		0.133		0.133
CAC 11kV Bus	CAC 11kV Bus	4000		2.062	4.202		0.052		0.075		1.112		0.133		0.133
CAC 11kV Line	CAC 11kV Line	4500		2.727	5.921		0.053		0.074		1.112		0.133		0.133
	EG – 33kV	2500	Site Specific (see Appendix 1.2)	1.063	2.596		0.058		0.073	Site Specific (see Appendix 1.2)	1.112		0.133		0.133
	EG – 11kV	3000		2.727	5.921		0.053		0.074		1.112		0.133		0.133
SAC Demand	HV Demand	8000	39.08		9.948	0.071				10.12	2.251	0.947			
	Large Demand	8100	18.86		13.718	0.071				10.12	2.251	0.947			
	Small Demand	8300	4.76		15.429	0.071				10.12	2.251	0.947			
SAC Non-demand	Business Flat	8500/8600 ^b	0.37			8.469				0.24		1.619			
	Business ToU	8800/8700 ^c	0.37				6.922		8.471	0.24			1.461		1.788
	Solar PV	9900	Government mandated tariff paid by Energex to the customer												
	Residential Flat	8400	0.28			8.728				0.07		1.472			
	Residential ToU	8900	0.28				6.309	9.566	19.771	0.07			1.187	1.803	3.754
	Super Economy (controlled load 1)	9000				3.475						0.637			
	Economy (controlled load 2)	9100				6.302						1.154			
	Unmetered	9600				6.836						1.252			

a. All tariffs are GST exclusive.

b. Customers on Network Tariff Code 8600 will be transitioned to NTC 8500 by 31/12/12.

c. Customers on Network Tariff Code 8700 will be transitioned to NTC 8800 by 31/12/12.



7 Tariff classes – alternative control services

For the 2010-15 regulatory control period the AER has classified fee-based services, quoted services and street lighting, as ACS. These three tariff classes are described in Table 7.1.

Table 7.1 Tariff classes for ACS

Tariff Class	Definition
Street Lighting	Street lighting services relate to activities of provision, construction and maintenance of street light assets owned by Energex (conveyance of electricity for street lights remains a SCS).
Fee-based Services	Services relating to activities undertaken by Energex at the request of customers or their agents (e.g. Retailers or contractors). The costs for these activities can be directly attributed to customers and service-specific charges can therefore be levied.
Quoted Services	Services for which the nature and scope cannot be known in advance irrespective of whether it is customer requested or an external event that triggers the need (e.g. price on application or compensable).

7.1 Street lighting

This section describes Energex's approach to the provision of the ACS of street lighting for the 2010-15 regulatory control period.

7.1.1 Overview

Since 1 July 2010, the bundled prices that customers are charged for street lighting consist of a SCS and an ACS. The conveyance of electricity to street lights remains a SCS, while services relating to the provision, construction and maintenance of street lighting assets have been classified by the AER as ACS. It is this ACS element of street lighting services that is discussed below.

In the 2010-15 regulatory control period, Energex will continue to provide for non-contributed and contributed street lighting services. Since 1 July 2010, the non-contributed service only applies where Energex has constructed standard street lights. Customers will still receive an ongoing charge for the standard level of service provided under the non-contributed service. Non-standard street lights will be available as a contributed service or as a quoted service, charged as an incremental cost to a standard street light. Charges associated with these services will need to be paid upfront by the customer. In the instances where work is required outside of business hours due to maintenance access restrictions or customer requirements, these services will be provided as a quoted service.



The AER approved a price cap form of control for street lighting services over the 2010-15 regulatory control period. This consisted of:

- a schedule of fixed prices for street lighting services for 2011-12; and
- a price path for the remaining years of the 2010-15 regulatory control period.

7.1.2 Framework

The approach for the treatment of street lighting assets, contributed and non-contributed, satisfies the requirements of the *Rules* and delivers network charges which directly correlate with the level of service provided.

Street lighting services where Energex constructs the asset

In the 2010-15 regulatory control period where a non-standard street light is requested by the customer, a separate charge for the incremental cost difference will be levied as a quoted service. The customer will still receive an ongoing charge for the non-contributed service.

Street lighting services where customer or agent constructs the asset

In the 2010-15 regulatory control period Energex will recognise gifted assets as contributed assets and will not seek to recover any asset related costs from the customer. The customer will receive an ongoing charge for the maintenance of the street light under the contributed service.

Existing Contributed

Contributed street lights existing pre 1 July 2010, will be recognised within SCS. This aligns with the historical capital contributions treatment where benefits from the street light asset contributions were accrued by all SACs. This reflects that all SACs are the end-beneficiary of the contributed street lighting service.

7.1.3 Charging parameters

Clause 6.18.2(b)(3) of the *Rules* states that:

A pricing proposal must set out, for each proposed tariff, the charging parameters and the elements of service to which each charging parameter relates.

Energex's revenue requirements for street lighting services have been determined based on the revenue building block components consistent with the approach used for its SCS. Based on the revenue requirement produced from the Post Tax Revenue Model (PTRM), Energex developed prices for the first year of the regulatory control period using a methodology that applies a revenue allocation based on the relative installation costs for major and minor street lights and the applicable asset funding arrangement (non-contributed and contributed).



Street lights are allocated to major and minor according to luminaire type and size, and to non-contributed and contributed based on the funding arrangement.

7.1.4 Tariffs

Clause 6.18.2(b)(2) of the *Rules* states that:

A pricing proposal must set out the proposed tariffs for each tariff class.

The Final Determination from the AER requires:

Energex to provide for street lighting, the proposed price for each service in the *regulatory year* and the revenues collected from the provision of each service in the preceding *regulatory year*.

Energex has provided a price schedule for the provision, construction and maintenance of street lights for the 2012-13 regulatory year in Appendix 2.1. The prices are based on the methodology approved in the AER Final Determination and charges are tailored to enable the customer to be charged according to the level of service requested. The prices reflect standardised lights and no restriction on access for operation, maintenance and repair. An additional charge may apply in the case of restricted access.

In line with the AER requirements, the revenues collected from the provision of each service in the preceding regulatory year are provided in Appendix 2.5.

The applicable terms and conditions for each street light service are outlined in Energex's published Tariff Schedule.

7.2 Fee-based and quoted services

This section describes Energex's approach to the provision of fee-based and quoted services for the 2010-15 regulatory control period.

7.2.1 Overview

Energex's fee-based services and quoted services are ancillary to the main distribution services and are provided at the explicit request of third parties.

- **Fee-based services** – are services relating to activities undertaken by Energex at the request of customers or their agents (e.g. Retailers or contractors). The costs for these activities can be directly attributed to customers and service-specific charges can therefore be levied.
- **Quoted services** – are services for which the nature and scope cannot be known in advance irrespective of whether it is customer requested or there is an external event that triggers the need.

For the 2010-15 regulatory control period, the prices for fee-based and quoted services will be subject to an annual escalation process which will take into account a weighted average price increase for each of the components of the price formula.



7.2.2 Framework

ACS are provided on an individual fee-for-service basis to Retailers and end use customers. Depending on the service, Energex will either provide the service for a fixed fee or quoted price.

The AER has approved a formula based price cap form of control mechanism for fee-based services and quoted services over the 2010-15 regulatory control period. This consists of:

- a schedule of fixed prices for fee-based services and a schedule of rates for quoted services for the first year of the 2010-15 regulatory control period; and
- an annual escalation process for the remaining years of the 2010-15 regulatory control period.

This formula outlined below for fee-based services and quoted services has been designed to ensure prices will be representative of the efficient costs of providing and delivering the service.

From 1 July 2010, large customer connections were classified as a quoted service under ACS. Additional detail on the framework and approach for large customer connections is provided in Section 7.3.

7.2.3 Charging parameters

Clause 6.18.2(b)(3) of the Rules states that:

A pricing proposal must set out, for each proposed tariff, the charging parameters and the elements of service to which each charging parameter relates.

In calculating the price for fee-based and quoted services, Energex will utilise the formula outlined below which includes cost parameters for different services to determine the price.

Price = Labour + Contractor Services + Materials + Capital Allowance + GST

- **Labour** (including overheads) – consists of all labour costs directly incurred in the provision of the service, labour on-costs, fleet on-costs and overheads. The labour cost for each service is dependent on the skill level, travel time, number of hours and crew size required to perform the service;
- **Contractor services** (including overheads) – reflects all costs associated with the use of external labour in the provision of the service, including overheads and any direct costs incurred as part of performing the service (e.g. traffic control, road closure permits);
- **Materials** (including overheads) – reflects the cost of materials directly incurred in the provision of the service, material storage and logistics on-costs and overheads; and
- **Capital allowance** – represents a return on and return of capital for non-system assets used in the delivery of the service.



7.2.4 Tariffs

Clause 6.18.2(b)(2) of the Rules states that:

A pricing proposal must set out the proposed tariffs for each tariff class.

The Final Determination from the AER requires:

Energex to provide, for fee-based and quoted services, the proposed price for each service in the *regulatory year*, in addition to the volume of services provided and revenues collected from the provision of each service in the preceding *regulatory year*.

Prices for fee-based services have been calculated using the above formula and forecast costs for labour, fleet on-costs, overheads and capital allowance.

The proposed price schedule for fee-based services in 2012-13 is provided in Appendix 2.2. The prices were determined using the AER's approved formula based price cap control mechanisms. Revised cost escalators and on-cost and overhead rates were calculated in accordance with the Final Determination. This has resulted in a price path for fee-based services which is slightly above the AER indicative price path outlined in the 2010 Final Determination. Additional information on the calculations and rates used is included in Appendix 2.4 (confidential).

The proposed price schedule for quoted services (illustrative configuration) in 2012-13 is provided in Appendix 2.3. These prices were determined using the AER's approved formula based price cap control mechanisms. The AER's approved prices do not represent a binding capped price for an individual quoted service.

Energex has retained its current policy of not establishing a fixed price where variations in the precise nature of the services being sought mean that averaging would result in significant inequity for customers. The prices for quoted services will be calculated using the formula above to reflect the actual cost of service provision based on the specific requirements of the customer.

In line with the AER requirements, the volume of services and the revenues collected from the provision of each fee-based and quoted service in the preceding regulatory year are provided in Appendix 2.5.



7.3 Large customer connections

7.3.1 Overview

From 1 July 2010, for new or upgraded connections (requested by a customer) of greater than 1 MVA or 4 GW.h pa (that is a tariff class of EG, CAC or ICC), the design and construction of the connection assets is a quoted service and will be priced as detailed in Section 7.2.3.

As part of the new service classification, customers may choose to have the service provided by other suppliers subject to those suppliers complying with Energex's Quality Assurance and materials specification requirements.

The commissioning, operation and maintenance of all connection assets, including large connections, was determined by the AER to be a SCS which represents no departure from the previous regulatory approach. In addition, the existing pricing methodology arrangements for the use of the shared network are unchanged.

7.3.2 Framework

The framework for new and existing large customer connections is provided in Figure 7.1 (classification of service) and Table 7.2 (pricing framework).

The design and construction of large customer connections will be classified as one of the following:

- **alternative control service** – for all new connections or upgrades to existing connections, which are paid for by the customer and gifted to Energex. This may include an upfront payment for the design and construction of the connection asset, calculated in accordance with the formula in Section 7.2.3. These assets will form part of the Contributed Asset Base (CAB). Items in the CAB will have no return on capital or regulatory depreciation cost allocated to them. However there will be an allocation for operating and maintenance costs (O&M) recovered through DUOS as per the tariff cost allocation process.
- **standard control service** – applies to large customer connections assets, existing prior to 1 July 2010, which are owned and maintained by Energex, or for Energex driven asset replacement. These services will continue to form part of the Regulatory Asset Base (RAB). The methodology of calculating network charges for these customers is consistent with the previous regulatory control period. Accordingly, these connection assets have costs allocated to them for return on capital, regulatory depreciation and O&M as per the tariff cost allocation process detailed in Appendix 3.
- **unregulated service** – applies to connection assets which are funded, owned and operated by the customer. These services will attract no specific connection asset charges.



Figure 7.1 Service classification for large customer connections

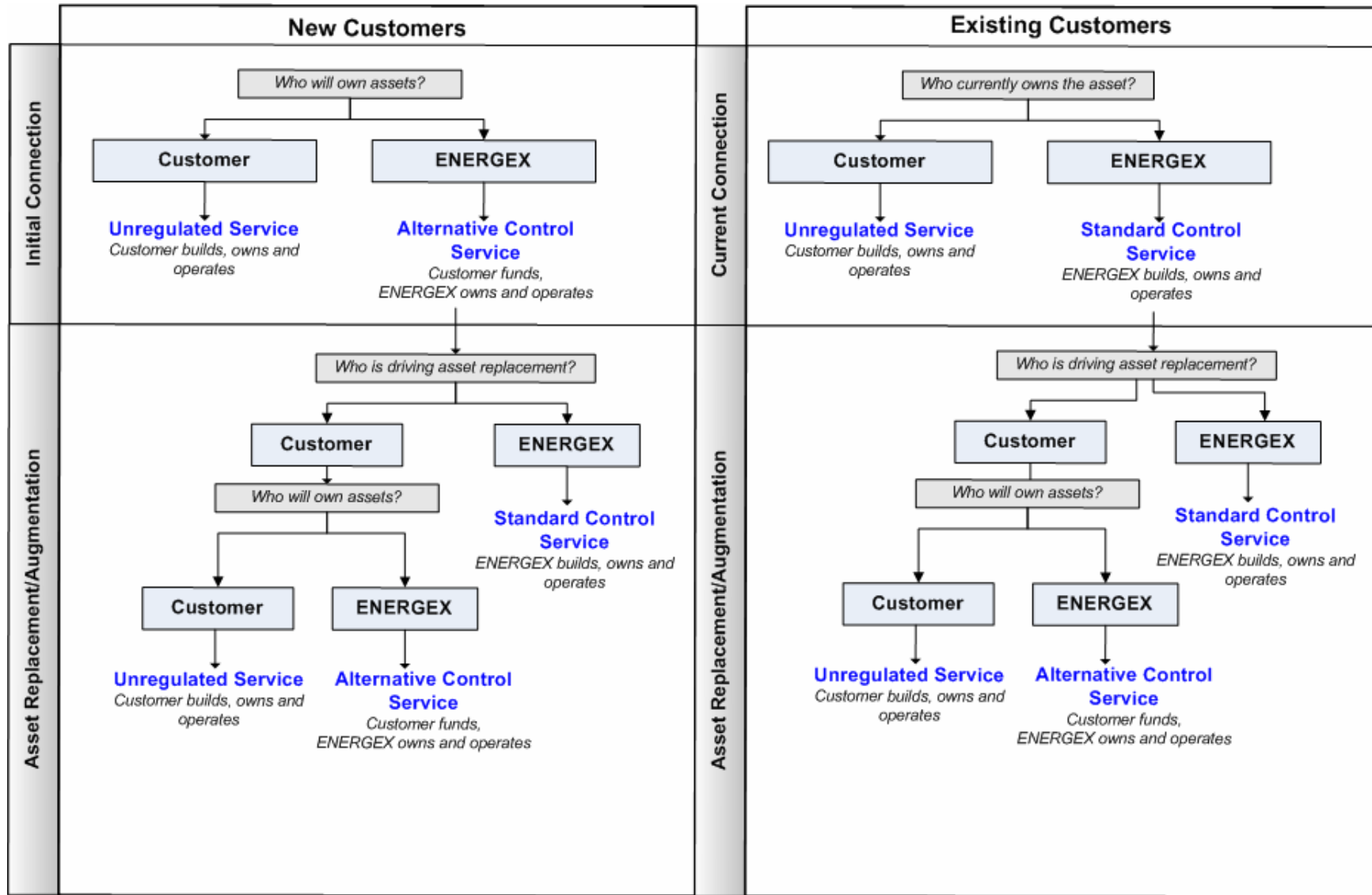


Table 7.2 Large customer connections pricing framework

Initial Connection Date	Description	Ownership ¹	Asset Classification	Service Classification	Asset Base	Upfront Customer payment (quoted price, relating to design and construction costs)	Tariff Charging Parameter (Site Specific Fixed Charge)	
							Recovery of depreciation and return on capital (through DUOS)	Recovery of operating expenditure (through DUOS)
Prior to 1 July 2010 (or part of transitional arrangement) (asset constructed under previous framework)	Existing connection	Energex	Non-contributed	SCS	RAB	n/a	✓	✓
	Upgrade (Energex driven) ²	Energex	Non-contributed	SCS	RAB	n/a	✓	✓
	Upgrade (customer request)	Energex (gifted)	Contributed	ACS	CAB	✓	n/a	✓
Post 1 July 2010 (asset constructed under new framework)	New Connection	Energex (gifted)	Contributed	ACS	CAB	✓	n/a	✓
	Upgrade (customer request)	Energex (gifted)	Contributed	ACS	CAB	✓	n/a	✓
	Upgrade (Energex driven) ²	Energex	Non-contributed	SCS	RAB	n/a	✓	✓
Asset constructed under either framework	Replacement (during warranty period for gifted assets)	Energex (gifted)	Contributed	n/a	CAB	n/a (covered under warranty)	n/a	✓
	Replacement (outside of manufacturers warranty period)	Energex (gifted)	Contributed	ACS	CAB	✓	n/a	✓
		Energex	Non-contributed	SCS	RAB	n/a	✓	✓
	Any service	Customer	n/a	Unregulated	n/a	No specific connection asset charges		

Standard Control Service (SCS), Alternative Control Service (ACS)

Regulatory Asset Base (RAB), Contributed Asset Base (CAB)

Notes

1. If the customer chooses to retain ownership of the asset, the service is unregulated and there are no specific connection asset charges.
2. An Energex driven upgrade to a customer's connection assets could occur, when for network reasons, the connection arrangement needs to be altered.



7.3.3 Charging parameters

Energex will calculate the price for the design and construction of large customer connection assets in accordance with the formula outlined in Section 7.2.3.

7.3.4 Tariffs

Clause 6.18.2(b)(2) of the *Rules* states that:

A pricing proposal must set out the proposed tariff for each tariff class.

Due to the variability in the scope, the design and construction of large customer connections is a quoted service for which no specific prices are published. The price for the illustrative service configuration for large customer connections is shown in Appendix 2.3.



8 Proposed tariff variations

8.1 Variations within the regulatory year 2012-13

SAC customer tariffs typically remain unchanged within the regulatory year. However, some SAC customers may participate in voluntary tariff trials during the regulatory year. These tariffs will be specifically designed to determine customer behaviour and may have special tariff elements for this purpose. More details on these tariffs are contained in Section 10.

The tariffs for larger customers are most likely to change where there are changes specific to an individual customer's connection arrangements and/or there is a material change in use of the site.

There is also the possibility of Government-mandated tariff changes. An example of such a change is the setting of prices for some of ACS via Schedule 8 of the *Queensland Electricity Regulation 2006*. Another example is the solar feed-in tariff, where a new tariff was introduced with pricing mandated by Government.

Any tariff pricing changes for either SCS or ACS during the 2012-13 regulatory year would be developed taking into account the same principles and objectives outlined in Section 4.1 of this proposal.

8.2 Variations between regulatory years 2011-12 and 2012-13

Energex has recently conducted a review of the existing network tariffs for standard asset customers (SAC). The various changes proposed for 2012-13, as outlined below, are designed to:

- support more specific pricing signals for residential customers through the introduction of a voluntary TOU tariff structure which more adequately reflects the cost of network augmentation thereby assisting in the management of peak demand;
- provide a more robust suite of network tariffs to aid customer understanding and simplify tariff application
- removal of fixed charges for super economy and economy tariffs; and
- enable market efficiency under the new Network plus Retail (N+R) methodology by supporting a 1:1 mapping of network tariffs to notified tariffs.



Table 8.1 Tariff Changes for 2012-13

Old Tariff	Revised Tariff
New	8900 – Residential TOU
8500 – Business Small – Flat 8600 – Business Medium - Flat ¹	Combined into one tariff: 8500 – Business Flat
8700 – Business Small – TOU 8800 – Business Medium - TOU ²	Combined into one tariff: 8800 – Business TOU
8200 – Demand Medium	Network tariff removed – all existing customers transferred to Demand Small or Demand Large.

¹ Customers on Network Tariff Code 8600 will be transitioned to NTC 8500 by 31/12/12.

² Customers on Network Tariff Code 8700 will be transitioned to NTC 8800 by 31/12/12.



9 Transmission cost recovery

Clause 6.18.2(b)(6) of the *Rules* states that:

A *pricing proposal* must set out how designated pricing proposal charges are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those charges in the previous regulatory year.

Clause 6.18.7 of the *Rules* states that:

- (a) A *pricing proposal* must provide for tariffs designed to pass on to customers the designated pricing proposal charges to be incurred by the *Distribution Network Service Provider* for transmission use of system services.
- (b) The amount to be passed on to customers for a particular *regulatory year* must not exceed the estimated amount of the designated pricing proposal charges adjusted for over or under recovery in accordance with paragraph (c).
- (c) The over and under recovery amount must be calculated in a way that:
 - (1) subject to subparagraphs (2) and (3) below, is consistent with the method determined by the AER in the relevant distribution determination for the *Distribution Network Service Provider*;
 - (2) ensures a *Distribution Network Service Provider* is able to recover from customers no more and no less than the designated pricing proposal charges it incurs; and
 - (3) adjusts for an appropriate cost of capital that is consistent with the rate of return used in the relevant distribution determination for the relevant regulatory year.
- (d) Notwithstanding anything else in this clause 6.18.7, a *Distribution Network Service Provider* may not recover charges under this clause to the extent these are:
 - (1) recovered through the *Distribution Network Service Provider's* annual revenue requirement;
 - (2) recovered under clause 6.18.7A; or
 - (3) recovered from another *Distribution Network Service Provider*.

The Final Determination from the AER requires:

Energex, in accordance with clause 6.12.1(19) of the *Rules*, to maintain a record of the amount of revenues recovered from TUOS charges and associated payments in accordance with Appendix E of the Final Determination.

In accordance with clauses 6.18.2(b)(6) and 6.18.7 of the *Rules*, tariffs outlined in this pricing proposal will allow for the pass through of DPPC, including any adjustments for under or over recovery. To comply with the *Rules*, information reported as part of this pricing proposal includes:



Expenses:

- regulated DPPC paid to Transmission Network Service Providers (TNSPs);
- avoided charges for the locational component of prescribed TUOS Services (to be referred to as avoided TUOS); and
- payments made to other DNSPs for use of their network.

Revenue:

- payments received from network users; and
- payments received from other DNSPs.

Adjustments for under/over recovery:

- difference between revenue and expenses.

9.1 Expenses

9.1.1 Transmission charges paid to TNSPs

Energex connects to the Powerlink network at multiple connection points. Powerlink, as a regulated TNSP, recovers its revenue from directly connected customers and DNSPs connected to its network.

In accordance with the connection agreement with Powerlink, Energex is required to pay DPPC to Powerlink on a monthly basis.

9.1.2 Avoided customer TUOS charges

Clause 5.5(h) of the *Rules* requires Energex to remit the locational component of prescribed TUOS services that would have been payable by Energex had the Connection Applicant not been connected to the network ('avoided charges for the locational component of prescribed TUOS services').

For embedded generators (EG), in accordance with the *Rules*, where prices for the locational component of prescribed DPPC services were in force at the relevant Transmission Network Connection Point (TNCP) throughout the relevant financial year, Energex shall:

- (a) determine the charges for the locational component of prescribed DPPC services that would have been payable by Energex had the EG not injected any energy at its connection point during that financial year;
- (b) determine the amount by which the charges calculated in (a) exceed the amount for the locational component of prescribed DPPC services actually payable by Energex; and
- (c) credit the value from (b) to the EG account.

In 2012-13, avoided TUOS payments will generally be remitted in the form of a lump sum payment after 30 June 2012, in line with previous years.



Avoided TUOS to EGs by Energex, reflect the avoided costs of upstream transmission network reinforcement to SEQ. As such, the benefits primarily relate to all customers – that is, avoided TUOS does not solely impact on the TNCP to which the EG is connected. Avoided TUOS has therefore been assigned to all tariff classes.

9.1.3 Payments to other DNSPs

In contingency circumstances, Essential Energy provides supply from its Terranora Substation in northern NSW to Energex's Kirra Zone Substation. Under this arrangement, Essential Energy requires Energex to pay for the use of its assets.

The charges established by Essential Energy in respect of this arrangement are based on approved rates for each month in which the alternate supply is utilised. Based on these charges, it is expected the total estimated charges for Essential Energy in 2012-13 will amount to \$805,000. These costs have been incorporated into the TUOS costs for the Mudgeeraba TNCP and therefore result in the charges from the use of Essential Energy network being passed through to users.

9.2 Revenue

9.2.1 Recovery of DPPC through tariffs

A description of how DPPC are recovered through standard control tariffs is given in Section 6.2. The total revenue received is indicated in Table 9.1.

9.2.2 Payments received from other DNSPs

Energex does not currently receive any transmission related payments from other DNSPs.

9.3 DPPC unders and overs account

As part of the requirements of the Final Determination, the AER requires Energex to provide amounts for the following entries in its DPPC unders and overs account for the most recently completed regulatory year (t-2) and the next regulatory year (t):

1. The opening balance for each year. The opening balance for year t-2 should be zero.
2. The amount of revenue recovered from DPPC applied in respect of that year, less the amounts of all transmission related payments made by Energex in respect of that year.
3. An interest charge for two years related to the net amount in item 2 for year t-2. This adjustment should be calculated using the approved nominal weighted average cost of capital (WACC).
4. The total of items 1 to 3 to derive the closing balance for each year.

Table 9.1 outlines the DPPC unders and overs calculation and provides separate identification of any under or over recovery relating to prior years included in the current year revenue. This distinction allows correct calculation of the under / over recovery of the respective year.



Table 9.1 Calculation of DPPC unders and overs account

	2010-11 actual (\$'000)	2012-13 forecast (\$'000)
Revenue from DPPC charges	326,673	405,224
Less under/over adjustment approved for year t-2	8,496	n/a
Total transmission related payments	343,772	394,867
Transmission charges to be paid to TNSPs	342,242	393,452
Avoided Customer TUOS payments	425	610
Inter – DNSP payments	1,105	805
Under/over recovery for regulatory year	-8,603	10,357
DPPC Unders & Overs Account		
Nominal WACC 2010-11	9.72%	n/a
Opening Balance	0	-10,357
Under/over recovery for regulatory year	-8,603	10,357
Interest on under/over recovery for regulatory year	-1,754	n/a
Closing Balance	-10,357	0

The AER requires that the amounts used in the table above for the most recently completed regulatory year (t-2) are audited.

Energex believes this requirement has been met as the figures are either:

- (a) audited as part of the statutory or regulatory account audits undertaken by Queensland Audit Office (QAO);
- (b) provided by the AER in the Final Determination; or
- (c) calculated based on underlying figures which are included under points (a) and/or (b) above.

Amounts for the next regulatory year (t) are forecast amounts.

In proposing variations to the amount and structure of DPPC for a given regulatory year t, Energex will achieve a zero expected balance on the DPPC unders and overs account at the end of each regulatory year in the regulatory control period.



10 Tariff development

Energex undertakes a continual program to improve the structure of its tariffs. A summary of the proposed changes to the structure of current tariffs or potential new tariffs to be introduced is provided below.

10.1 Structural changes to tariffs

10.1.1 kV.A tariffs for large customers

Establishment of the kV.A tariff

Following proactive engagement with stakeholders on the issue of kV.A, Energex introduced kV.A tariffs for ICC and CAC customers from 1 July 2010.

Network charges for ICC and CAC customers are calculated using the existing kW based approach. The demand and capacity charges are adjusted up or down by a relative amount based on the difference between the customer's measured power factor and the power factor neutral point. This methodology is described further in Appendix 1.3.

Rationale

From Energex's perspective, providing the correct pricing signals will assist in ensuring that the utilisation of its network is at its most efficient – an outcome which is of benefit to all customers, as it will reduce the need to spend additional funds on increasing the capacity of the network, thereby minimising tariff increases.

The two methods of measuring utilisation of the network, or customer demand, are kW or kV.A. The power factor, or kV.A, is important because transmission and distribution systems must be designed and built to manage the need for real power as well as its associated reactive component – the total power. The total power can be greater than 50 per cent more than real power alone if the power factor is poor. Ideally, power factor should be as close to unity as possible. Therefore the further it moves away from one, the more network capacity is needed to supply the same kW demand. This is demonstrated in Table 10.1 below (where total power (kV.A) = demand (kW) / power factor).

Table 10.1 Impact of power factor on the network

	Demand (kW)	Power Factor	Total Power (kV.A)
Customer 1	1,600	0.7	2,286
Customer 2	1,600	0.8	2,000
Customer 3	1,600	0.9	1,778
Customer 4	1,600	1.0	1,600



A tariff based on kV.A is therefore a more accurate measure of a customer's impact on the network, relative to a tariff based on kW, as it better reflects the costs imposed on the network by users. Accordingly, a kV.A tariff better meets Energex's pricing principles as it is more cost-reflective and encourages more efficient use of the network.

Further development of the kV.A tariff

The power factor adjustment methodology outlined above is a transitional method to help customers understand the concept of kV.A and the impact of kV.A on the network and network charges. Energex intends to move to a straight kV.A tariff, reflective of current practices in other Australian jurisdictions, from July 2013.

For 2012-13, the power factor adjustment will be increased to provide the equivalent kV.A price. This will provide an additional incentive for customers with a poor power factor to improve and is more cost reflective.

The kV.A methodology will be revenue neutral for the ICC tariff class and across the total of all CAC tariff classes.

10.1.2 Tariffs for Residential Customers

From 1 July 2012, in addition to the default flat rate tariff for residential customers, the network tariff structure will include a voluntary ToU tariff.

Residential Time of Use (ToU) Tariff

This tariff requires a multi-function electronic meter that is capable of supporting ToU tariffs. The residential ToU tariff is structured with different rates for peak, shoulder and off-peak periods, as outlined in Section 6. The objective of a ToU energy charge is to reduce the demand on the network during peak times by encouraging customers to switch non-essential electricity use to off-peak or shoulder periods.

10.2 Tariff Trials and EC&DM Programs

Energex is implementing a number of tariff trials and Residential programs as part of a strategy to encourage more efficient usage of the network. These programs and associated customer research will provide Energex with information on potential customer behaviour with respect to new tariffs.

10.2.1 Controlled load tariffs

Controlled load is load that can be managed at Energex's discretion within specific periods as agreed with the customer. Controlled load provides a means to manage peak demand during times of maximum network utilisation. This reduces the capacity requirements on the network. Load may be interrupted without loss of utility benefits to customers or, where customers volunteer, some utility benefit may be forgone for a financial and/or environmental reward.



Air-conditioning Demand Management

Domestic and commercial air-conditioning load continues to be a significant contributor to peak demand. During the 2010-15 regulatory control period, Energex is continuing to build on the early success of its 'time for a cool change' (Cool Change) trials, by rolling out Peak Smart Air-Conditioning broadly across Energex's distribution area from September 2012. This work supports the new demand management technology being used by several air-conditioning manufacturers (previously limited to Energy Conservation Communities program only). As part of this program, a signal receiver will be supplied for customers who wish to purchase a new Peak Smart air-conditioning unit. Effective incentives are critical to the ongoing success of the Peak Smart program and appropriate tariffs need to be developed to reward customers who demand manage their air-conditioning units.

Pool pump direct load control

The Cool Change and Energy Conservation Communities programs also include trials focused on providing load control solutions for pool pumps. Estimates indicate pool pumps contribute significantly to system peak demand and initial analysis has indicated that benefits exist in shifting pool pump usage from peak use times. Accordingly, in addition to offering and promoting a better rate for electricity used by pool pumps at non-peak times through tariff 33, Energex will continue to offer alternative customer choice in relation to pool pump load control where tariff conversion may not be practical or feasible.

Energy Conservation and Demand Management has recently launched a broad based pool program (with heavy pool industry and electrician support) rebating customers who either connect their pool pump to Tariff 33 or convert a fixed speed pool pump to an 5 star rated (or better) energy efficient pool pump. To date, over 3,000 customers have applied for this broad based offer with a target of 9,000 proposed for 2012-13 including a broad based advertising program to drive these offers.

Hot water load control

In regard to hot water load control, Energex is currently conducting a campaign across SEQ to offer incentives to convert from a continuous supply tariff (retail notified tariff 11) to a less expensive tariff that provides supply for a minimum eight hours (super economy– retail tariff 31) or 18 hours (economy– retail tariff 33) per day. In addition, Energex is reviewing switching times with the view of optimising the amount of load managed during summer peaks. A rewards campaign is currently being shaped to run through the 2012-13 winter period to incentivise customers to take advantage of tariff conversions. A rough target of 3,000 customers is proposed for the 2012-13 hot water conversion offer.



10.2.2 Reward-based tariff (RBT) trial

The Reward Based Tariffs trial is a cooperative project between Energex and Ergon Energy to trial how electricity prices may be used to encourage customers to alter their electricity consumption. Currently in its second year, the trial involving approximately 3,700 customers in selected suburbs throughout Cairns, Toowoomba and Brisbane is due to run until 31 December 2012.

Two types of tariffs are being tested throughout the trial, including a time varying tariff (incorporating a dynamic peak price and a time-of-use price) forming the consumption trial; and a maximum demand (capacity based) tariff, specifically targeting high demand days. The trial will not change the way customers are billed by Retailers, rather they receive performance statements that provide feedback on their performance based on the trial's theoretical electricity prices. Customers are therefore encouraged to respond to these theoretical trial prices through reward incentives.

The trial aims to gain an understanding of customer acceptance and behaviour towards tariffs of this nature and develop Energex's understanding of the effectiveness of such tariffs in reducing peak demand. Information obtained through the trial will feed into future tariff policy development.

10.3 Future Tariff Changes

10.3.1 ToU demand for large customers

Energex's network tariffs include a demand component for all customers greater than 100 MW.h per annum.

The demand component is currently based on the maximum demand recorded in the month (Any Time). Demand tariffs signal to customers that they can reduce their network charges by reducing the variability of their consumption.

If a ToU signal was introduced in combination with the demand signal this would provide greater cost reflectivity and encourage more efficient use of the network. This tariff structure would also reward large customers who are able to shift demand to off-peak periods as a lower peak demand reduces the pressure that they place on the network.

Energex will consider the potential introduction of a ToU element for demand tariffs as part of its longer term pricing strategy.

10.3.2 Appliance tariff

Domestic air-conditioning load continues to be significant contributor to summer peak demand. The option to control air-conditioning, or other appliances, already exists under an existing load control tariff which provides a regular on/off switching program. However this requires a separately metered circuit and requires the appliances to be switched completely off during the peak period making this unattractive to most customers.



Energex has successfully trialled cycling of air-conditioning in the Cool Change Program using an Energex signal receiver fitted to the air-conditioner unit. The trials have demonstrated that this method of controlled demand reduction does not impact customer comfort.

Energex is currently considering the option of an appliance tariff which would reward customers who install an Energex signal receiver on their home appliances. The customer benefit would reflect the cost of deferred capital expenditure on the network.

This tariff could be used for other major appliances such as pool pumps as the technology becomes available.

10.4 Proposed removal of tariffs

Energex does not anticipate the removal of any tariffs during the 2013-14 regulatory year.



11 Compliance with regulatory requirements

11.1 DUOS unders and overs account

The Final Determination from the AER requires:

Energex to keep and provide information on a distribution use of system (DUOS) unders and overs account under clause 6.18.2(b)(7) of the *Rules*.

As part of the requirements of the Final Determination, the AER requires Energex to provide entries in its DUOS unders and overs account for the most recently completed regulatory year (t-2) and the next regulatory year (t).

In accordance with Section 4 of the Final Determination, where Energex has an under or over recovery balance compared to the MAR which is between two and five percent, the adjustment to revenue may span two years. Where the under/over recovery is not cleared in one year, additional interest is charged on the remaining balance to index it to the year in which it is adjusted.

In 2010-11 the closing balance on the unders and overs account is \$20.5 million. This represents 1.3% when compared to the MAR.



Table 11.1 Calculation of DUOS unders and overs account

	2010-11 actual (\$'000)	2012-13 forecast (\$'000)
Revenue from DUOS charges	1,029,442	1,491,186
Less under/over adjustment approved for year t-2 (from t-4)	-24,400	-
Reversal of foregone revenues	-	-45,511
Less MAR for the relevant year	1,108,870	1,516,239
Allowed revenues (AR_t)	1,135,064	1,468,944
Service quality performance reward/penalty (S_t)	-	29,379
Capital contribution overs/unders adjustment (C_t)	902	824
Transitional adjustments ($Transitional_t$)	-27,095	-
Approved pass throughs ($Pass-through_t$)	-	17,093
Under/over recovery for regulatory year	-55,028	20,457
DUOS unders and overs account		
Nominal WACC 2010-11	9.72%	n/a
Opening balance – 2010/11	38,034	-20,458
Interest on opening balance	7,753	n/a
Under/over recovery for regulatory year	-55,028	20,457
Interest on under/over recovery for regulatory year	-11,217	n/a
Closing Balance	-20,458	0

Note: individual components may not sum to the total due to rounding

The AER requires that the amounts used in the table above for the most recently completed regulatory year (t-2) are audited.

Energex believes this requirement has been met as the figures are either:

- (a) audited as part of the statutory or regulatory account audits undertaken by Queensland Audit Office (QAO);
- (b) provided by the AER in the Final Determination; or
- (c) calculated based on underlying figures which are included under points (a) and/or (b) above.

Amounts for the next regulatory year (t) are forecast amounts.



11.2 Compliance with avoidable and stand alone cost requirements

Clause 6.18.5 of the *Rules* states that:

- (a) For each *tariff class*, the revenue expected to be recovered should lie on or between:
- (1) an upper bound representing the stand alone cost of serving the customers who belong to that class; and
 - (2) a lower bound representing the avoidable cost of not serving those customers.

11.2.1 Standard control services

In line with the *Rules*, the revenue expected to be recovered from each tariff class will lie on or between the bounds of stand alone and avoidable costs. This requirement ensures that the revenue recovered from each tariff class is cost reflective and free from cross subsidies.

Section 11.2.1 describes how this requirement is met, while Table 11.3 provides estimates which indicate that the expected revenue to be recovered from tariffs for 2012-13 is between the two bounds.

Definitions

Stand alone costs are the costs that would be incurred to service a group of customers (referred to as a tariff class) on a stand alone basis.

- Stand alone costs can be estimated in two ways. Firstly, the cost can be estimated by assuming that a new hypothetical network is constructed solely for the particular tariff class. Secondly, the costs can be estimated by calculating the cost of using the existing network to supply the particular tariff class on a stand alone basis. For the purposes of this document, Energex will use the second method. It is noted that due to economies of scale, the total stand alone costs based on the first method will be greater than the figure calculated using this second method. However, as this is the basis upon which the expected revenue (and subsequent tariffs) is calculated and the estimated stand alone cost represents the upper bound, this is considered to be a satisfactory method for this purpose.
- Avoidable costs reflect the costs that would be avoided if a particular group of customers (referred to as a tariff class) were no longer connected to the network, assuming that all other customers remain connected.

Background

Energex calculates tariffs using a Distribution Cost of Supply (DCOS) model. The DCOS model generates DUOS tariffs based on the full distribution of the building block costs approved by the AER. Appendix 3 outlines the tariff cost allocation process which is used by Energex.

The DCOS model has also been used to estimate the stand alone and avoidable costs for each tariff class. Table 11.2 outlines which of the DCOS cost categories are used in the calculation of the two price boundaries.



In addition, the conversion of the DCOS cost categories to the tariff charging parameters is provided in Table 11.2.

Table 11.2 DCOS categories used in price boundary calculations and the conversion of DCOS categories into tariffs

DCOS Cost Category		Pricing Boundaries		Tariff Charging Parameter		
		Avoidable cost	Stand alone cost	Fixed or Service Availability Charge (\$/period)	Volume Charge ² (c/kW.h)	Capacity / Demand Charge (\$/kW)
Network O&M	Non-contributed Connection Assets	✓	✓	✓		
	Contributed Connection Assets	✓	✓	✓		
	Network Assets		✓		✓ ¹	✓
Network Regulatory Depreciation	Non-contributed Connection Assets	✓	✓	✓		
	Contributed Connection Assets ³					
	Network Assets		✓		✓ ¹	✓
Network Return on Capital	Non-contributed Connection Assets	✓	✓	✓		
	Contributed Connection Assets ³					
	Network Assets		✓		✓ ¹	✓
Common Services			✓		✓	
Non-System			✓	✓	✓	✓

1 Applicable to customers on volume based network tariffs.

2 The structure of the volume charge can incorporate either a flat rate or a Time of Use volume charge.

3 There is no regulatory depreciation or return on capital for contributed connection assets.

Lower bound test (avoidable cost)

As shown in Table 11.2, the avoidable costs for a tariff class include the cost of connection and the costs of operating and maintaining that connection. The fixed or standing charge for the customer includes these costs, making it the floor price. Therefore, any use of the shared network will incur additional charges (in the form of the volume and/or capacity/demand charge parameters), taking the charge paid by any customer above the avoidable cost of supply (the economic cost floor).



Upper bound test (stand alone cost)

In the DCOS model the infrastructure, operating and maintenance costs for upstream assets are shared across multiple customers and tariff classes. For this reason, the allocated cost of supply for each tariff class will be equal to, or below the stand alone costs of supply.

In the case of smaller network customers connected at the distribution level (generally 11kV and below), the allocated cost will be well below stand alone costs as the costs for high voltage assets are shared with larger customers.

For larger network customers connected at the sub-transmission level (33kV and above), the allocated cost model includes a site-specific parameter of supply network costs. The cost-based tariffs for CACs take into account the specific connection costs as well as an allocation of the upstream shared network costs. ICC cost-based tariffs are determined by mapping the actual supply network and allocating the relevant proportion of costs to the customers on the basis of their use of that network.

Therefore as there is an allocation of costs, and/or the full network costs are allocated in the case of a single-user asset, the revenue recovered from tariffs must be equal to or below the stand alone cost of supply (economic cost ceiling).

Estimates

The methodology used by Energex in calculating stand alone and avoidable costs is described below.

The avoidable cost estimate for each tariff class has been developed assuming:

- Avoidable capital - the annual capital charge is taken as the return on capital and depreciation allocations for the non-contributed connection assets required to service the tariff class.
- Avoidable O&M - the annual O&M costs are taken as those costs allocated to all connection assets for each tariff class in the DCOS model. It is assumed that the shared O&M costs for common and non-system assets are not avoidable as they are incurred irrespective of whether one particular tariff class is no longer connected.
- Total avoidable costs - are a sum of the above and are represented as an annual charge in Table 11.3 below.

The stand alone cost estimate for each tariff class has been developed assuming:

- The existing network is used to supply the particular tariff class on a stand alone basis.
- Stand alone capital - the annual capital charge is taken as the return on capital and depreciation allocations for the non-contributed connection assets and all network (shared) assets required to service the tariff class.
- Stand alone O&M - the annual O&M costs are taken as the total allocated for each tariff class in the DCOS model. This includes costs for connection and network assets, common services and non-system assets.
- Total stand alone costs - are a sum of the annual capital and O&M charges above and are represented as an annual charge in Table 11.3 below.



Table 11.3 Stand alone and avoidable cost boundaries for 2012-13

Tariff Class	Avoidable cost (\$m)	Expected revenue (\$m) (excl. side constraint adjustment)	Stand alone cost (\$m)
ICC	9.8	32.4	33.4
CAC - 33kV	1.7	3.9	4.3
CAC - 11kV Bus	4.1	18.9	19.2
CAC - 11kV Line	6.5	85.3	87.5
EG	0.4	1.1	1.4
SAC Demand	22.7	310.6	343.4
SAC Non-demand	78.7	1,038.9	1,084.9

Note: The Expected Revenue in the table above for Clause 6.18.5 of the *Rules* excludes the adjustment to MAR resulting from the Government directive.

11.2.2 Alternative control services

The formula used for calculating prices for fee-based services and quoted services has been designed to ensure prices will represent the efficient costs of providing and delivering the service and signal the economic costs of service provision by being subsidy free. Prices based on this formula will be cost reflective, representing costs derived through the same allocative mechanism as that used to determine costs for SCS, in accordance with the AER approved Cost Allocation Method (CAM). The prices for each tariff class within ACS will be between the bounds of avoidable and stand alone costs due to the economies of scale in providing each service.

The avoidable cost for a particular service is equivalent to the direct labour and materials cost. Overhead costs will be incurred regardless of whether or not the service is provided. Consequently, the price calculated using the formula in Section 7.2.3 (which includes an allowance for overhead costs) will be greater than the avoidable cost.

The stand alone cost is equal to the costs of serving each tariff class within ACS on a stand alone basis. For example, the stand alone cost would require the use of dedicated resources and assets (including costs associated with corporate support, capital allowance for non-system assets, etc.). As these costs can be shared among all of the tariff classes within ACS, the cost calculated for each individual service will be less than the stand alone cost.



11.3 Long run marginal cost

Clause 6.18.5(b)(1) of the Rules states that:

A tariff, and if it consists of 2 or more *charging parameters*, each *charging parameter* for a *tariff class*:

- (1) must take into account the long run marginal cost for the service or, in the case of a *charging parameter*, for the element of the service to which the *charging parameter* relates.

11.3.1 Standard control services

Marginal cost is the change in total cost that arises when the quantity produced changes by one unit. In the case of an electricity network, the marginal cost could be the cost incurred from one additional customer connecting to the network, or an additional MW of demand consumed.

Marginal costs can be calculated as either short run marginal costs (SRMC) or long run marginal costs (LRMC). In the short run, investment in capacity is fixed; therefore, the SRMC refers to the cost of a customer connecting to the network but using only the existing capacity in the network. In the long run, investment in capacity is variable; hence LRMC indicates an estimate of the cost of connecting the customer when an augmentation to the capacity of the network is necessary. Pricing on the basis of LRMC therefore assumes that prices should be based on the cost of meeting an increase in demand over an extended period of time.

In the electricity market, as demand on the network increases, an expansion of network capacity is required to accommodate the additional demand. In order to provide correct signals for customers, prices must take into account these additional capital costs. The LRMC should be considered in setting tariffs as it can provide appropriate signalling to customers and will assist in achieving a stable tariff structure over the forecast period.

One method which is used to estimate the LRMC is the Average Incremental Cost (AIC) method:

$$\text{LRMC (AIC)} = \frac{\text{PV(Capex)} + \text{PV(Opex)}}{\text{PV(Incremental Demand)}}$$

where, *PV (Capex) and PV (Opex) represent the present value (PV) costs associated with meeting the additional demand over the regulatory control period.*

This calculation method gives rise to an estimate which smoothes out irregular capital expenditure over time, whilst providing an indication of the costs associated with the increased demand. Thus, the LRMC indicates the level at which future increments of output must be priced to ensure total cost recovery given forecast demand.

The incremental capital and operating costs associated with increased demand are included in the Energex forecast capital and operating expenditure programs for 2010-15. Energex receives a return on these (and other) costs in the form of a return on capital and depreciation



charges as approved by the AER. These costs are then allocated to each tariff class as part of the DCOS model.

Energex designs tariffs to contain a combination of charging parameters in order to reflect LPMC and recover the total allowable revenue:

- where appropriate, a capacity charge may take into account the long term demand peak and can provide effective pricing signals to customers of the next increment of load;
- a demand charge is used to take into account short term peaks in demand;
- volume charges are used where appropriate due to limitations with current metering; and
- fixed charges are used to ensure the remaining costs including the costs associated with connection assets are recovered.

11.3.2 Alternative control services

Prices for ACS are calculated based on the current costs to provide a particular service. The LPMC is taken into consideration when determining prices for ACS. However, given that there is only a single charge parameter for each alternative control service, prices have been set at average cost. This ensures total costs are recovered for these services, as allowed for under clause 6.18.5(c).

11.4 Transaction costs

Clause 6.18.5(b)(2)(i) of the *Rules* states that:

A tariff, and if it consists of 2 or more *charging parameters*, each *charging parameter* for a *tariff class*:

- (2) must be determined having regard to:
 - (i) transaction costs associated with the tariff or each *charging parameter*.

For each tariff, Energex has selected a number of charging parameters (identified in Section 6.2 for SCS and Section 7.1.3 and 7.2.3 for ACS). Each combination of charging parameters in a tariff has been selected to adequately reflect the need for both fixed and variable components.

Energex has sought to simplify pricing structures without sacrificing economic functionality, whilst also minimising transaction and pricing administration costs.

A combination of various parameters is required in order to achieve economic functionality and to ensure that appropriate pricing signals are provided to customers. However, the number and design of these parameters has been selected with regard to minimising the associated transaction costs.



11.5 Response to price signals

Clause 6.18.5(b)(2)(ii) of the Rules states that:

A tariff, and if it consists of 2 or more *charging parameters*, each *charging parameter* for a *tariff class*:

- (2) must be determined having regard to:
 - (ii) whether customers of the relevant *tariff class* are able or likely to respond to price signals.

11.5.1 Standard control services

Consistent with the *Rules*, the tariffs proposed by Energex are comprised of a number of charging parameters that provide signals to customers about the efficient use of the network and the impact of future network capacity and costs. For SCS, the charging parameters and their respective cost signals are discussed in detail in Section 6.2.

The different types of charging parameters used in network pricing are:

- **capacity charges** – signals to network customers that they can reduce their network charges by reducing the variability of their consumption, as this lower peak demand reduces the pressure that they place on the network;
- **demand charges** – signals to network customers that they can reduce their charges by reducing their maximum demand, as it is levied on the basis that network customers who place greater pressure on the system should incur higher charges; and
- **volume charges** – by using a Time of Use structure (where metering allows) a volume charge can provide a small price signal that encourages more efficient use of the network by moving load from peak to off-peak times.

Energex has designed these parameters with pricing signals that customers are able to respond to. Energex is aware of customers putting systems in place to manage their network charges in response to these signals, for example, commercial and industrial customers (who have these elements as part of the network tariff) may stagger the starting time of equipment such as motors, air-conditioning and even large lighting installations. This has the effect of reducing their maximum demand on the network, as all the start-up loads are not occurring at the same time. The short-term benefit to the customer is a lower monthly demand charge and if they effectively manage their longer term any time maximum demand (ATMD) they will benefit from reduced capacity charges.

Energy usage costs are affected by the overall electrical efficiency of installations. If customers improve the efficiency of their usage this will reduce their energy related charges. For example, this can be achieved by installing more efficient lighting and improved thermal insulation of cold rooms and air-conditioned premises.



11.5.2 Alternative control services

Prices for ACS are calculated based on the current costs to provide a particular service. The pricing signals present are restricted to differentiating between services performed during business hours and those performed after hours. Customers are able to minimise the costs incurred for these services by choosing to have them delivered during business hours, where possible.

11.6 Tariff adjustment to address revenue shortfalls

Clause 6.18.5(c) of the *Rules* states that:

If, however, as a result of the operation of paragraph 6.18.5(b), the *Distribution Network Service Provider* may not recover the expected revenue, the provider must adjust its tariffs so as to ensure recovery of expected revenue with minimum distortion to efficient patterns of consumption.

When setting its network tariff, Energex uses a combination of charging parameters. These are developed taking into account the LRMC, transaction costs and customer response to pricing signals, as required under 6.18.5(b).

However, the expected revenue would not be recovered if charging parameters considered only those items identified in 6.18.5(b). Accordingly, other parameters are used in order to allow for the shortfall in expected revenue. These parameters are selected in a manner which complements the chosen pricing signals and minimises distortion.

11.7 Compliance with side constraints

Clause 6.18.6 of the *Rules* states that:

- (a) This clause applies only to *tariff classes* related to the provision of *standard control services*.
- (b) The expected weighted average revenue to be raised from a *tariff class* for a particular *regulatory year* of a *regulatory control period* must not exceed the corresponding expected weighted average revenue for the preceding *regulatory year* by more than the permissible percentage.
- (c) The permissible percentage is the greater of the following:
 - (1) the CPI-X limitation on any increase in the *Distribution Network Service Provider's* expected weighted average revenue between the two *regulatory years* plus 2%;
Note: The calculation is of the form $(1 + \text{CPI})(1 - X)(1 + 2\%)$
 - (2) CPI plus 2%.
Note: The calculation is of the form $(1 + \text{CPI})(1 + 2\%)$



- (d) In deciding whether the permissible percentage has been exceeded in a particular *regulatory year*, the following are to be disregarded:
- (1) the recovery of revenue to accommodate a variation to the distribution determination under rule 6.6 or 6.13;
 - (2) the recovery of revenue to accommodate pass through of designated pricing proposal charges to customers; and
 - (3) the recovery of revenue to accommodate pass through of jurisdictional scheme amounts for approved jurisdictional schemes.
- (e) This clause does not, however, limit the extent a tariff for customers with remotely-read interval metering or other similar metering technology may vary according to the time or other circumstances of the customer's usage.

In the annual pricing proposal, Energex is required to demonstrate that the proposed DUOS tariffs for the next year (t) will meet the below side constraint formula for each tariff class. This formula and requirement is set by the AER in the Final Determination.

$$\frac{\sum_{j=1}^m d_t^j \times q_t^j}{\sum_{j=1}^m d_{t-1}^j \times q_t^j} \leq (1 + \Delta CPI_t) \times (1 - X_t) \times (1 + 2\%) \pm S_t \pm C_t \pm \text{transitional}_t \pm \text{passthrough}_t \pm \text{unders and overs}_t$$

where each tariff class ' j ' has up to ' m ' components, and where:

d_t^j is the proposed price for component j of the tariff class for year t

d_{t-1}^j is the price charged by Energex for component j of the tariff class in year $t-1$

q_t^j is the forecast quantity of component j of the tariff class in year t

ΔCPI_t is the annual percentage change in the ABS Consumer Price Index All Groups, Weighted Average of Eight Capital Cities from March in regulatory year $t-2$ to March in regulatory year $t-1$.

X_t is the X factor for each year of the regulatory control period. If $X > 0$, then X will be set equal to zero for the purposes of the side constraint formula.

S_t is the Service Target Performance Incentive Scheme (STPIS) factor to be applied in regulatory year t .

C_t is the annual adjustment factor for the difference between actual and forecast capital contributions in year $t-2$.

transitional_t is a transitional factor for matters such as under/over in tax paid during the current regulatory control period and under/over adjustments related to shared assets used for purposes other than SCS.

pass-through_t is an annual adjustment factor that reflects the pass through amounts approved by the AER with respect to regulatory year t .

$\text{unders \& overs}_t$ is an annual adjustment factor related to the balance of the DUOS under and over account with respect to regulatory year t .



In compliance with Clause 6.18.6(d) of the *Rules*, as shown by the formula above, the permissible percentage excludes all DPPC charges and an adjustment is made for any approved pass-throughs.

Table 11.4 demonstrates Energex's compliance with the side constraint formula for each tariff class in 2012-13.

Energex has applied the side constraint to all tariff classes (with the exception of ICCs and EGs) in 2012-13. The side constraint results in a net impact on revenue of \$8.6 million (as shown in Figure 4.1).

Where Energex has imposed a side constraint for 2012-13 tariffs and it is identified that an actual under recovery occurs, Energex will recover the under recovered revenues only from that tariff class that is under recovered and not across all tariff classes.

The calculation of the permissible percentage change shown below is based on the CPI from March 2011 to March 2012 (All Groups, Weighted Average of Eight Capital Cities) of 1.58%.

Table 11.4 Compliance with side constraint formula

Tariff Class	Weighted Average Revenue 2011-12 (\$m) (regulatory year t-1)	Expected Revenue 2012-13 (\$m) (regulatory year t, incl. side constraint adjustment)	% change (calculated)	% change (permissible)
ICC	28.6	32.4	13.07%	20.44%
CAC - 33kV	3.5	3.9	13.14%	20.44%
CAC - 11kV Bus	16.3	18.8	15.45%	20.44%
CAC - 11kV Line	70.6	82.8	17.37%	20.44%
EG	1.0	1.1	14.09%	20.44%
SAC Demand	258.2	304.7	17.98%	20.44%
SAC Non-demand	880.0	1,038.9	18.06%	20.44%

In order to calculate the percentage change for each tariff class, the weighted average revenue for 2011-12 in Table 11.4 is based on 2011-12 rates and 2012-13 quantities, as specified in the side constraint formula.



11.8 Summary of compliance with the *Rules* and distribution determination requirements

Clause 6.18.2(b)(7) of the *Rules* states that:

A pricing proposal must demonstrate compliance with the Rules and any applicable distribution determination.

Table 11.5 provides the pricing proposal reference to meet the requirements of each clause of Section 6.18 of the *Rules*.



Table 11.5 Demonstrated compliance with the *Rules*

Clause	Pricing Proposal Requirement	Reference
6.1.4(a)	Demonstrates that Energex does not charge a Distribution Network User DUOS charges for the export of electricity generated by the user into the distribution network.	Section 6.1
6.1.4(b)	Demonstrates that where a Distribution Network User generates and exports electricity into the distribution network, Energex charges for the provision of connection services.	Section 6.1
6.18.2(b)(1)	Sets out each tariff class (including the classes of alternative control services).	Section 1 Section 7
6.18.2(b)(2)	Sets out the proposed tariffs for each tariff class.	Table 6.5 Appendix 1 Appendix 2
6.18.2(b)(3)	Set out for each proposed tariff, the charge parameters and the elements of service to which each charging parameter relates.	Section 6.2 Section 7.1.3 Section 7.2.3
6.18.2(b)(4)	Set out, for each tariff class related to standard control services, the expected weighted average revenue for the relevant regulatory year and also for the current regulatory year.	Section 11.7
6.18.2(b)(5)	Sets out the nature of any variation or adjustment to the tariff that could occur during the course of the regulatory year and the basis on which it could occur.	Section 8.1
6.18.2(b)(6)	Set out how designated pricing proposal charges are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those charges in the previous regulatory year.	Section 6.2.5 Section 9
6.18.2(b)(6A)	Set out how jurisdictional scheme amounts for each approved jurisdictional scheme are to be passed on to customers and any adjustments to tariffs resulting from over or under recovery of those amounts.	Section 4.2.1
6.18.2(b)(6B)	Describe how each approved jurisdictional scheme that has been amended since the last jurisdictional scheme approval date meets the jurisdictional scheme eligibility criteria.	
6.18.2(b)(7)	Demonstrates compliance with the <i>Rules</i> and any applicable distribution determination.	Section 11 Table 11.5 Table 11.6
6.18.2(b)(8)	Describes the nature and extent of change from the previous regulatory year and demonstrates that the changes comply with the <i>Rules</i> and any applicable distribution determination.	Section 8 Section 12
6.18.3(a)	Define the tariff classes into which customers for <i>direct control services</i> are divided.	Section 5 Section 6.1 Section 7



Clause	Pricing Proposal Requirement	Reference
6.18.3(b)	Demonstrates that each customer for <i>direct control services</i> is a member of at least one tariff class.	Section 5
6.18.3(c)	Sets out separate tariff classes for standard control and alternative control customers.	Section 5 Section 6.1 Section 7
6.18.3(d)(1)	Demonstrates that tariff classes are formed based on groupings of customers on an economically efficient basis.	Section 5
6.18.3(d)(2)	Demonstrates that customers and tariffs are grouped into tariff classes with regard to the need to avoid unnecessary transactions costs.	Section 11.4
6.18.4(a)(1)(i), (ii) and (iii)	Demonstrates that customers are assigned (or re-assigned) to tariff classes on the basis of the nature and extent of their usage, the nature of their connection to the network, and the metering installed at the customer's premises.	Section 5
6.18.4(a)(2)	Demonstrates that customers with a similar usage and connection profile are treated on an equal basis.	Section 5
6.18.4(a)(3)	Demonstrates that customers with micro-generation facilities are treated no less favourably than customers without such facilities.	Section 5.1
6.18.4(a)(4), (b)	Demonstrates that customer assignment (or reassignment) to a tariff class does not occur in the absence of an effective system of assessment and review.	Section 5.1.1
6.18.5(a)(1) and (2)	Demonstrates that revenue from a tariff class lies between the stand alone and avoidable cost.	Section 11.2
6.18.5(b)(1)	Demonstrates that tariffs and tariff components have regard for long run marginal cost.	Section 11.3
6.18.5(b)(2)(i)	Demonstrates that tariffs and tariff components have regard for the transaction costs to customers.	Section 11.4
6.18.5(b)(2)(ii)	Demonstrates that tariffs and tariff components are set with regard to whether customers are able or likely to respond to price signals.	Section 11.5
6.18.5(c)	Demonstrates that if tariffs do not recover the required revenue as a result of the operation of clause 6.18.5(b), the tariffs have been adjusted with minimum distortion to efficient patterns of consumption.	Section 11.6
6.18.6(b)	Demonstrates that the weighted average revenue for a standard control tariff class does not exceed that for the previous year by more than the "permissible percentage" defined in clause 6.18.6(c) of the <i>Rules</i> .	Section 11.7
6.18.6(c)(1) and (2)	Demonstrates the "permissible percentage" has been calculated in accordance with the definition set out in this clause of the <i>Rules</i> .	Section 11.7
6.18.6(d)(1) and (2)	Demonstrates that Designated Pricing Proposal Charges and pass-throughs were removed from the calculation of the side constraint.	Section 11.7



Clause	Pricing Proposal Requirement	Reference
6.18.6(e)	Demonstrates that the side constraints have not impacted on the extent to which the tariffs for a customer with remotely read interval metering will vary according to usage.	Section 11.7
6.18.7(a)	Demonstrates that tariffs passed on to customers include the charges to be incurred by Energex for <i>transmission use of system services</i> .	Section 6.2 Section 9
6.18.7(b)	Demonstrates that the DPPC charges passed on to customers do not exceed the estimated DPPC charges adjusted for over or under recovery in the previous regulatory year.	Section 9.3
6.18.7(c)(1) and (2)	Demonstrates that any DPPC over or under recovery is the difference between what was paid to Powerlink and what was recovered from customers via DPPC charges.	Section 9.3
6.18.9(a)(1)	The tariff classes and the applicable tariffs (or prices) are published on Energex's website.	The 2012-13 tariff schedule will be published on the ENERGEX website in June 2011.
6.18.9(a)(2)	The charging parameters within each tariff class (i.e. the fixed, demand and energy prices) and the elements of the service to which each charging parameter relates are published on Energex's website.	
6.18.9(a)(3)	The manner in which tariffs are expected to change over the remainder of the regulatory control period, and the reasons for the changes, are published on Energex's website.	The 2012-13 statement of price trends will be published on the ENERGEX website in June 2011.
6.18.9(b)	The information set out in clause 6.18.9(a) is published on ENERGEX's website 20 days prior to the start of the relevant regulatory year or as soon as practicable thereafter.	

Table 11.6 provides the pricing proposal reference for other specific requirements under the Final Determination.



Table 11.6 Demonstrated compliance with the requirements of the Final Determination

Section	Pricing Proposal Requirement	Reference
Section 2.5.2	Energex to follow the procedure for assigning customers to tariff classes or reassigning customers from one tariff class to another as specified in Appendix B of the Final Determination.	Section 5.1.1
Section 4.5.2	Demonstrates that the proposed DUOS prices for the next year (t) will meet the side constraints formula set out in Section 4.5.2 of the Final Determination.	Section 11.7
Section 4.6	Include a record of the amount of revenues recovered from TUOS charges and associated payments in accordance with Appendix E of the Final Determination.	Section 9.3
Section 17.3.3	Include the revenues collected from the provision of each street light service in the preceding regulatory year.	Appendix 2.5
Section 18.3.3	Calculate the respective internal and external labour cost escalators for each regulatory year of the next regulatory control period and provide both qualitative and quantitative supporting information to the AER as part of the annual pricing proposals	Appendix 2.4
Section 18.3.4	Calculate the on cost and overhead rates in each regulatory year of the next regulatory control period in accordance with the methodologies set out in the CAM and submit both qualitative and quantitative supporting information to the AER as part of the annual pricing proposal.	Appendix 2.4
Section 18.3.5	Include the volume of fee-based and quoted services provided and the revenues recovered from the provision of fee-based and quoted services in the preceding regulatory year.	Appendix 2.5
Section 18.3.5	Set out the nature and extent of any variation to an individual formula component, on cost or overhead rate from that applicable in the previous regulatory year that is above the indicative illustrative quoted service examples and fee-based services set out in Appendix I, J and K of the final decision.	Appendix 2.4
Section 18.3.5	Set out in its pricing proposal the nature and extent of any variation or adjustment to the methodology employed to derive a formula component escalator or on cost or overhead rate.	Appendix 2.4
Appendix D	Maintain a distribution use of system (DUOS) unders and overs account. Energex must provide information on this account to the AER as part of its annual pricing proposals.	Section 11.1
Appendix E	Maintain a transmission use of system (TUOS) unders and overs account. Energex must provide information on this account to the AER as part of its annual pricing proposals.	Section 9.3



12 Customer impacts

12.1 2012-13 regulatory year

12.1.1 Standard control services

Energex is cognisant of the changing expectations of customers and the current upward pressure being exerted on energy prices and has considered this when developing its pricing strategy. As an organisation Energex is committed to achieving a balanced commercial outcome between meeting the requirements of customers and managing sustainability and risk.

The ongoing average price increases across the regulatory control period are necessary for Energex to deliver the capital and operating expenditure programs which support development and growth in SEQ, as well as improving reliability and security of supply.

Table 12.1 provides an estimate of the charges in 2012-13 for the average consumption level in each tariff class. This table is used to give an estimate of the percentage change for the average customer. The impact on each customer will be dependent on the individual customer's demand and consumption patterns.

Table 12.1 Outline of estimated % price change by tariff class for 2011-12 to 2012-13

Tariff Class	Estimated DUOS charge 2011-12 (c/kW.h)	Estimated DUOS charge 2012-13 (c/kW.h)	Percentage change (%)
ICC	1.6	1.8	13.07%
CAC – 33kV Line/Bus	2.9	3.3	13.14%
CAC – 11kV Bus	1.9	2.2	15.45%
CAC – 11kV Line	2.5	2.9	17.37%
EG	1.8	2.0	14.09%
SAC Demand	4.9	5.8	17.98%
SAC Non-demand	7.9	9.4	18.06%

12.1.2 Alternative control services

The price changes between 2011-12 and 2012-13 for ACS (street lighting and fee-based services) are provided in Table 12.2 and Table 12.3.



Table 12.2 Street light services % increase for 2011-12 to 2012-13

Street lighting service	% change (2011-12 to 2012-13)
Major non-contributed	13.98%
Major contributed	12.06%
Minor non-contributed	15.33%
Minor contributed	11.68%

Table 12.3 Fee-based services % increase for 2011-12 to 2012-13

Fee-based Service	% change (2011-12 to 2012-13)
Alterations and additions to current metering equipment	7.40%
Attending loss of supply – LV customer installation at fault (BH)	7.40%
Overhead service replacement – single phase	7.40%
Overhead service replacement – multiple phase	7.40%
De-energisation	7.40%
Meter test	7.40%
Meter inspection	7.40%
Reconfigure meter	7.40%
Off-cycle meter read	7.40%
Site visit	7.40%
Locating Energex underground cables	n/a
Temporary connection	7.40%
Re-energisation – business hours	7.40%
Re-energisation – after hours	7.40%
Re-energisation (visual) – business hours	7.40%
Re-energisation (visual) – after hours	7.40%
Re-energisation non-payment (visual) – business hours	7.40%
Re-energisation non-payment (visual) – after hours	7.40%
Supply abolishment	7.40%
Unmetered supply	7.40%
Street light glare screening	7.40%
Replacement of standard luminaries with aero screen units (per street light)	7.40%



12.2 Expected price trends (2011-12 to 2014-15)

Clause 6.18.9(a)(3) states that Energex must maintain on its website:

a statement of expected price trends (to be updated for each *regulatory year*) giving an indication of how the *Distribution Network Service Provider* expects prices to change over the *regulatory control period* and the reasons for the expected changes.

The statement of the expected price trends for 2011-12 to 2014-15 is published annually on the Energex website.

In summary:

- for SCS, any change in prices will be subject to the approved X factors from the Final Determination, the change in CPI and any yearly adjustments which are applicable;
- in each regulatory year (as part of the requirements of a pricing proposal under the *Rules*), Energex will demonstrate that the proposed tariffs for DUOS will meet the side constraints formula for each tariff class; and
- for ACS including street lighting and fee-based services, the prices will be subject to an annual escalation process.



13 Annual approval process

Energex will submit annually to the AER a schedule of proposed tariffs that are to apply in the following financial year. Energex will demonstrate that the proposed tariffs comply with the *Rules* and the requirements of the AER Final Determination.

For the 2012-13 annual pricing proposal, the dates for the annual approval process are:

30 April 2012	Energex submits to the AER, at least 2 months before the commencement of the second and each subsequent regulatory year of the regulatory control period, a further pricing proposal (an "annual pricing proposal") for the relevant regulatory year.
31 May 2012	Approval from the AER if satisfied that the proposed prices comply with the <i>Rules</i> and the requirements under the determination.
June 2012	Energex publishes price schedule for the coming financial year on Energex's website 20 business days prior to the start of the relevant regulatory year or as soon as practicable thereafter.



14 Glossary

14.1 Abbreviations

Term	Definition
ACS	Alternative Control Service
AER	Australian Energy Regulator
AIC	Average Incremental Cost
ATMD	Anytime Maximum Demand
c/kW.h	cents per Kilowatt hour
CAB	Contributed Asset Base
CAC	Connection Asset Customers
CAM	Cost Allocation Method
CPI	Consumer Price Index
DCOS	Distribution Cost of Supply
DNSP	Distribution Network Service Provider
DPPC	Designated Pricing Proposal Charges (previously known as TUOS)
DUOS	Distribution Use of System
EG	Embedded Generators
GW.h	Gigawatt hour
HV	High Voltage
ICC	Individually Calculated Customers
kV	Kilovolt
kV.A	Kilovolt Ampere
kW.h	Kilowatt hour
LCC	Large Customer Connection
LRMC	Long Run Marginal Cost
LV	Low Voltage
MAR	Maximum Allowable Revenue
MSATS	Market Settlement and Transfer Solution
MW	Mega Watt
NEL	National Electricity Law
NEM	National Electricity Market



NMI	National Meter Identifier
NMP	Network Management Plan
NTC	Network Tariff Code
NUOS	Network Use of System
O&M	Operating and Maintenance Allowance (Opex)
PFA	Power Factor Adjustment
PTRM	Post Tax Revenue Model
PV	Photovoltaic
PV	Present Value
QAO	Queensland Audit Office
QCA	Queensland Competition Authority
QESI	Queensland Electricity Supply Industry
RAB	Regulatory Asset Base
RBT	Reward-based tariff
RIN	Regulatory Information Notice
<i>Rules</i>	National Electricity Rules
SAC	Standard Asset Customers
SCI	Statement of Corporate Intent
SCS	Standard Control Services
SEQ	South East Queensland
SRMC	Short Run Marginal Cost
STPIS	Service Target Performance Incentive Scheme
TNCP	Transmission Network Connection Point
TNSP	Transmission Network Service Provider
TOU	Time of Use
TUOS	Transmission Use of System
WACC	Weighted Average Cost of Capital

14.2 Definitions

Term	Definition
After Hours (A/H)	Anytime outside of business hours
Alternative Control Service	This service class includes the provision, construction and maintenance of street lighting assets, and fee-based and quoted services.
Anytime (Any)	At the Retailer's request, the work is performed within business hours to speed up the completion, and, where necessary after hour fees are charged.
Anytime Maximum Demand (ATMD)	This is the maximum demand recorded at anytime. Can be used to indicate the maximum demand that occurs anytime within a specified time period such as a 12 calendar month period or a specific month.
Business Hours (B/H)	Monday to Friday 8am to 5pm.
Capacity charge	This part of the tariff seeks to reflect the costs associated with providing network capacity required by a customer on a long-term basis. It is levied on the basis of either contracted demand or the maximum demand in the previous calendar year. The charge is applied as a fixed dollar amount per kW per month.
Capacity - Network	The maximum demand (kW) that the distribution network can provide for at any one time.
Charging Parameter	The constituent elements of a tariff (as per the <i>Rules</i> definition).
Common Service	A service that ensures the integrity of a distribution system and benefits all Distribution Customers and cannot reasonably be allocated on a locational basis.
Connection Asset Customer (CAC)	Typically, those customers with electricity consumption greater than 4GW.h (but less than 40GW.h) per year at a single connection point; or where demand is greater than or equal to 1 MVA; or where a customer has a dedicated supply system with significant connection assets or the customer has contributed to their dedicated connection assets.
Connection Asset (Contributed or Non-contributed)	Related to building connection assets at a customer's premises as well as the connection of these assets to the distribution network. Connection assets can be contributed (customer funded, then gifted to Energex) or non-contributed (Energex funded).
Connection Point	The point of electrical coupling between the electricity distribution network and a customer's electrical installation. The meter is installed as close as possible to this location.
Controlled Load 1	Refer to Super Economy
Controlled Load 2	Refer to Economy
Demand	The amount of electricity energy being consumed at a given time measured in either watts (W) or volt amperes (VA). The difference between the two is the power factor.



Term	Definition
Demand Metered SAC	The customer's connection point has a meter installed that is capable of measuring energy consumption (kW.h) and demand (kW). This meter records total energy consumption (kW.h) and demand over 30 minute periods. A customer's demand is the average demand (kW) over the 30 minute period.
Demand Metered Tariff	The tariff has been structured to include a demand component so the customer's actual demand is reflected in the price they pay for their electricity. The highest demand reading for that month is used to calculate the customer's electricity bill.
Demand Charge	This part of the tariff accounts for the actual demand a customer places on the electricity network. The actual demand levied for billing purposes is the metered monthly maximum demand. The charge is applied as a fixed dollar per kW per month.
Distribution Cost of Supply (DCOS) Model	The Energex model used to allocate costs approved by the AER to the various tariff classes.
Distribution Loss Factor (DLFs)	These represent the average electrical energy losses incurred when electricity is transmitted over a distribution network.
Distribution Use of System (DUOS)	This refers to the network charges for the use of the distribution network.
Designated Pricing Proposal Charge (DPPC)	Refers to the charges incurred for use of the transmission network; previously referred to as Transmission Use of System (TUOS).
Economy (controlled load 2)	Specified connected appliances are controlled by network equipment so that supply will be available for a minimum period of 18hours per day during time periods set at the absolute discretion of Energex.
Embedded Generator (EG)	In line with the Energy Networks Association (ENA) classification, Embedded Generators (EGs) are generally those generators with an installed capacity as follows: Medium: 1-5 MW (LV or HV) or <1MW (HV) Large: > 5MW
Energy	The amount of electricity consumed by a consumer (or all customers) over a period of time. Energy is measured in terms of watt hours (W.h), kilowatt hours (kW.h), megawatt hours (MW.h) or gigawatt hours (GW.h).
Fixed Charge	The fixed charge seeks to reflect the costs associated with customer's dedicated connection assets. The charge is applied as a fixed dollar amount per day.
High Voltage (HV)	Refers to the 11kV or above network.
Individually Calculated Customer (ICC)	Typically those customers with electricity consumption greater than 40GW.h per year at a single connection point; or where the customers demand is greater than or equal to 10 MVA; or where a customer's circumstances mean that the average shared network charge becomes meaningless or distorted.
kV.A	Kilovolt Amperes, which is a measure of the apparent power being consumed and is used to measure demand. One kV.A equals 1,000 Volt Amperes.



Term	Definition
Low Voltage (LV)	Refers to the sub-11 kV network
Maximum Allowable Revenue (MAR)	The maximum revenue which can be recovered through tariffs for the regulatory year.
Maximum Demand	The maximum demand recorded at a customer's individual meter or the maximum demand placed on the electrical distribution network system at any time or at a specific time or within a specific time period, such as a month. Maximum demand is an indication of the capacity required for a customer's connection or the electrical distribution network.
Micro Generator	AS4777 compliant generators with an installation size of less than 10kW (single phase) or 30kW (three phase) connected to the LV network.
Market Settlement and Transfer Solution (MSATS)	The central repository for Standing Data for all NMIs in contestable markets.
National Electricity Market (NEM)	The interconnected electricity grid covering Queensland, New South Wales, Victoria, Tasmania, South Australia and the Australian Capital Territory.
National Metering Identifier (NMI)	A unique number assigned to each metering installation.
Network Use of System (NUOS)	The tariff for use of the distribution and transmission networks. It is the sum of both Distribution Use of System (DUOS) and Transmission Use of System (TUOS).
Non-demand Metered SAC	The customer's connection point has a meter installed that is capable of measuring the total energy consumption (kW.h).
Non-demand Metered Tariff	The tariff is based around a fixed daily component and the actual energy (kW.h) used by the customer.
Off-peak Period	All hours which are outside of Peak Period and Shoulder hours.
Peak Period	Meter type 1-4 (ICC, CAC & SAC demand): The hours between 7am and 11pm, Monday to Friday. Meter type 6 (SAC Non-demand - Business): The hours between 7am and 9pm, Monday to Friday. Meter type 6 (SAC Non-demand - Residential): The hours between 4pm and 8pm, Monday to Friday.
Power Factor	Power factor, is the ratio of kW to kV.A, and is a useful measure of the efficiency in the use of the network infrastructure. The closer to one the power factor, the more efficient the network assets are utilised. Power Factor = kW/kV.A
Price Path	Outlines the escalation factors to be applied to the initial price over the regulatory control period.
Queensland Government Solar Bonus Scheme for Standard Asset Customers	A program that pays residential and other small energy customers for the surplus electricity generated from roof-top solar photovoltaic (PV) systems that is exported to the Queensland grid.



Term	Definition
Regulatory Depreciation	Also referred to as the return of capital—the sum of the (negative) straight–line depreciation and the (positive) annual inflation effect on the opening regulatory asset base (RAB).
Return on Capital	The return necessary to achieve a fair and reasonable rate of return on the assets necessarily invested in the business.
Shoulder Period	Meter type 6 (SAC Non-demand - Residential): The hours between 7am to 4pm and 8pm to 10pm, Monday to Friday and 7am to 10pm weekends.
Site Specific Charge	This charge is calculated specifically for a site and is specific to the individual connection point.
Small generator	AS4777 compliant generators with an installation size of less than 1MW connected to the LV network.
Solar Photovoltaic (Solar PV)	A system that uses sunlight to generate electricity for residential use. The system provides power for the premises with any excess production feeding into the electricity grid.
Standard Asset Customer (SAC)	Generally those customers with an annual electricity consumption below 4GW.h per year, whose supply arrangements are consistent across the customer group; and where there is no contribution for their dedicated connection assets.
Standard Control Service	This service class includes network, connection and metering services.
Street lights (Major)	Lamps in common use for Major Road lighting including: a) High Pressure Sodium 100 watt (S100) and above; b) Metal Halide 150 watt (H150) and above; and c) Mercury Vapour 250 watt (M250) and above.
Street lights (Minor)	All lamps in common use for Minor Road lighting, including Mercury Vapour, High Pressure Sodium and Fluorescent.
Super Economy (controlled load 1)	Specified permanently connected appliances are controlled by network equipment so that supply will be permanently available for a minimum period of 8 hours at the absolute discretion of Energex but usually between the hours of 10:00pm and 7:00am.
Tariff class	A class of customers for one or more <i>direct control services</i> who are subject to a particular tariff or particular tariffs (as per the <i>Rules</i> definition).
Time of Use (ToU)	Refers to tariffs that vary according to the time of day at which the electricity is consumed.
Transmission Use of System (TUOS)	Refers to the charges incurred for use of the transmission network.
Unmetered Supply	A customer who takes supply where no meter is installed at the connection point.



Term	Definition
Volume (Energy) Charge	This part of the tariff seeks to reflect costs not directly allocated to network drivers and costs that are proportional to the size of the customer. The energy consumption (kW.h) for the period, as recorded by the customer's meter, is utilised to calculate this part of the tariff charge. This charge is applied as a fixed amount (cents) per Kilowatt hour (kW.h).
Volume (Energy) Charge (Off-peak)	This charge is applicable to those customers who are on a Residential and Business, Time of Use tariff. The energy consumption (kW.h) during Off-peak periods (refer to Off-peak Period for Off-peak period times), as recorded by the customer's meter, is utilised to calculate this part of the tariff. This charge is applied as a fixed amount (cents) per Kilowatt hour (kW.h).
Volume (Energy) Charge (Peak)	This charge is applicable to those customers who are on a Residential and Business, Time of Use tariff. The energy consumption (kW.h) during Peak periods (refer to Peak Period for Peak period times), as recorded by the customer's meter, is utilised to calculate this part of the tariff. This charge is applied as a fixed amount (cents) per Kilowatt hour (kW.h).
Volume (Energy) Charge (Shoulder)	This charge is applicable to those customers who are on a Residential Time of Use tariff. The energy consumption (kW.h) during shoulder periods (refer to Shoulder Period for Shoulder period times), as recorded by the customer's meter, is utilised to calculate this part of the tariff. This charge is applied as a fixed amount (cents) per Kilowatt hour (kW.h).



15 Confidential information

Claim for confidentiality

Energex claims confidentiality over the following sections and appendices on the grounds that the information is either Commercial-in-confidence or contains intellectual property. Energex requests that the AER does not disclose the information contained in these sections to any person outside of the AER.

Appendices

No.	Title
1.1	ICC site specific tariffs
1.2	CAC and EG site specific tariffs
2.4	Price development for alternative control services
2.5	Additional reporting requirements for alternative control services



16 Appendices

No.	Title
1	2012-13 Tariffs – Standard Control Services 1.1 ICC site specific tariffs (Confidential) 1.2 CAC and EG site specific tariffs (Confidential) 1.3 kV.A calculation methodology for ICC and CAC customers
2	2012-13 Tariffs – Alternative Control Services 2.1 Street lighting services 2.2 Fee-based services 2.3 Quoted services – Representative sample 2.4 Price development for alternative control services (Confidential) 2.5 Additional reporting requirements for alternative control services (Confidential)
3	Tariff cost allocation process



Appendix 1

2012-13 Network Tariffs

Standard Control Services

Contents

- 1.1 ICC site specific tariffs (confidential)
 - 1.2 CAC and EG site specific tariffs (confidential)
 - 1.3 kV.A calculation methodology for ICC and CAC customers
-

Appendix 1.1

CONFIDENTIAL

2012-13 Tariffs Standard Control Services ICC

Energex will provide these site specific tariffs directly to the customer and their Retailer.



Appendix 1.2

CONFIDENTIAL

2012-13 Tariffs Standard Control Services CAC and EG Site Specific Tariffs

Energex will provide these tariffs directly to the customer and their Retailer.



Appendix 1.3

2012-13 Tariffs Standard Control Services kV.A calculation methodology for ICC and CAC customers

Commencing 1 July 2010, charges for ICC's and CAC's have been adjusted up or down by a relative amount based on the difference between the customers' measured power factor and the power factor neutral point. The adjustment will be applied to the demand and capacity charge elements for 2012-13 as outlined below.

1. kV.A Tariff Adjustment Calculation Methodology

a.
$$\text{Actual Power Factor (APF)} = \frac{\text{Maximum monthly kW}}{\text{Maximum monthly kV.A}}$$

where monthly kW and monthly kV.A data is collected from the customer's meter

b.
$$\text{Power Factor Adjustment (PFA)} = \text{Multiplier} \times (\text{TPF} - \text{APF}) \times (1 + \text{TPF} - \text{APF})^{\text{index}}$$

where TPF = Target Power Factor (Power factor neutral point of 0.9 for ICCs, 0.85 for CACs)

Multiplier and index are variables set by Energex (as per table below)

The PFA will be limited to the maximum and minimum bounds defined by Energex as required

c.
$$\text{Demand Adjustment (kW)} = \text{PFA} \times \text{Maximum monthly demand}$$

$$\text{Capacity Adjustment (kW)} = \text{PFA} \times \text{Nominated monthly demand}$$

The adjustments (measured in kW) will be positive or negative depending on the difference between the customers actual power factor (APF) and target power factor (TPF). This will result in an increase or decrease to the monthly charge as shown in the formula below.

d.
$$\text{Monthly charge for DUOS(\$)} = \underbrace{\text{Fixed} + \text{Demand} + \text{Capacity} + \text{Volume Peak} + \text{Volume Off Peak}}_{\text{Existing parameters}} + \underbrace{\text{Demand Adjustment} + \text{Capacity Adjustment}}_{\text{kV.A adjustments for 2011/12}}$$

where Demand Adjustment (\$) = PFA x Maximum monthly demand (kW) x Demand Charge (\$/kW/month)

Capacity Adjustment (\$) = PFA x Nominated monthly demand (kW) x Capacity Charge (\$/kW/month)

2. kV.A Tariff Adjustment Parameters for 2012-13

Customer Type	Target Power Factor	Index	Multiplier	Maximum Adjustment	Minimum Adjustment
ICC - 110B ¹	0.9000	1.0000	1.0000	0%	0%
ICC - 33B	0.8939	1.5138	1.0900	30%	-30%
ICC - 11B	0.8939	1.5138	1.0900	30%	-30%
CAC - 33L	0.8691	1.4306	1.1500	40%	-40%
CAC - 11B	0.8691	1.4306	1.1500	40%	-40%
CAC - 11L	0.8691	1.4306	1.1500	40%	-40%

¹ The power factor adjustment will not be applied to ICC 110 customers due to metering limitations



3. kV.A Tariff Adjustment Example

The following example is used to illustrate the application of the kV.A tariff adjustment for a customer of the CAC 11kV Line tariff class.

Input Parameter	Value	Reference
Tariff Class	CAC 11kV Line	
Nominated kW Demand	2000kW	
Maximum kW Demand	1800kW	Collected from customer's meter data
Maximum kVA Demand	2000kW	Collected from customer's meter data
Multiplier	1.1500	As specified for tariff class CAC 11 Line
Index	1.4306	As specified for tariff class CAC 11 Line
Maximum adjustment	40%	As specified for tariff class CAC 11 Line
Minimum adjustment	-40%	As specified for tariff class CAC 11 Line
Target Power Factor	0.8691	As specified for CAC tariff class
Demand Charge	\$5.921 /kW/month	As proposed for 2012-13 in Table 6.5 (DUOS only)
Capacity Charge	\$2.727 /kW/month	As proposed for 2012-13 in Table 6.5 (DUOS only)

a. Actual Power Factor (APF) = $\frac{\text{Maximum monthly kW}}{\text{Maximum monthly kV.A}} = \frac{1800}{2000} = 0.900$

b. Power Factor Adjustment (PFA) = $\text{Multiplier} \times (\text{TPF} - \text{APF}) \times (1 + \text{TPF} - \text{APF})^{\text{index}}$

$$= 1.15 \times (0.8691 - 0.90) \times (1 + 0.8691 - 0.90)^{1.4306}$$

$$= -0.03397$$

$$= -3.4\%$$

The calculated PFA is between the minimum and maximum adjustment bounds, so the limiting boundaries do not need to be applied.

c. Demand Adjustment (kW) = PFA x Maximum Monthly Demand (kW)

$$= -0.03397 \times 1800$$

$$= -61.15 \text{ kW}$$

Capacity Adjustment (kW) = PFA Nominated Monthly Demand (kW)

$$= -0.03397 \times 2000$$

$$= -67.94 \text{ kW}$$

d. Demand Adjustment (\$) = PFA x Maximum Monthly Demand (kW) x Demand Charge (\$/kW/month)

$$= -0.03397 \times 1800 \times \$5.921$$

$$= -\$362.04$$

Capacity Adjustment (\$) = PFA x Nominated Monthly Demand (kW) x Capacity Charge (\$/kW/month)

$$= -0.03397 \times 2000 \times \$2.727$$

$$= -\$185.27$$

Note: some rounding may be evident in the above calculations



Appendix 2

2012-13 Network Tariffs

Alternative Control Services

Contents

2.1	Street lighting services
2.2	Fee-based services
2.3	Quoted services
2.4	Price development for alternative control services (confidential)
2.5	Additional reporting requirements for alternative control services (confidential)

Appendix 2.1

2012-13 Tariffs Alternative Control Services Street lighting services

The proposed prices for the provision, construction and maintenance of street light services are provided in Table 1 below. The prices below are as approved in the May 2010 Final Determination.

Table 1 Energex prices for street lighting services

Streetlight Service	2012-13 Price (\$ per light per day)
Major non-contributed	1.01
Major contributed	0.27
Minor non-contributed	0.41
Minor contributed	0.11

All prices are exclusive of GST.

In accordance with the Energex Public Lighting Manual, the following definitions for major and minor lights apply:

- **Major Lamps** in common use for Major Road lighting mean the following:-
 - a) High Pressure Sodium 100 watt (S100) and above
 - b) Metal Halide 150 watt (H150) and above
 - c) Mercury Vapour 250 watt (M250) and above
- **Minor Lamps** are all lamps in common use for Minor Road lighting other than Major Lamps and include Mercury Vapour, High Pressure Sodium and Fluorescent.



Appendix 2.2

2012-13 Tariffs Alternative Control Services Fee-based services

The proposed prices for individual fee-based services provided by Energex are outlined in Table 1 below. These prices were determined using the AER's approved formula based price cap control mechanisms.

Table 1 Prices for fee-based services for the 2012-13 regulatory year

Fee-based Service	2012-13 Price ¹ (\$ per service)
Alterations and additions to current metering equipment	101.61
Attending loss of supply – low voltage customer installation at fault – business hours	113.50
Overhead service replacement – single phase	310.73
Overhead service replacement – multiple phase	366.04
De-energisation ²	50.89
Meter test ²	117.59
Meter inspection	90.82
Reconfigure meter	75.30
Off-cycle meter read ²	8.48
Site visit	65.22
Locating Energex underground cables	n/a
Temporary connection ²	903.50
Re-energisation – business hours ²	44.34
Re-energisation – after hours ²	126.54
Re-energisation (visual) – business hours ²	75.39
Re-energisation (visual) – after hours ²	165.32
Re-energisation non-payment (visual) – business hours ²	75.39
Re-energisation non-payment (visual) – after hours ²	165.32
Supply abolishment	351.93
Unmetered supply	155.32
Street light glare screening	150.11
Replacement of standard luminaries with aero screen units (per street light)	347.02

¹ All prices are exclusive of GST.

² These services are subject to Schedule 8 of the Queensland Electricity Regulation 2006. However, as the Schedule 8 prices for 2012-13 are not yet known, the above rates reflect proposed Energex cost using the alternative control services formula. The Schedule 8 prices will be included in the Energex 2012-13 Tariff Schedule, which will be published on the Energex website prior to 1 July 2012.



Appendix 2.3

2012-13 Tariffs Alternative Control Services Quoted services

The prices for the illustrative configuration of Energex's quoted services are outlined in Table 1 below. These prices were determined using the AER's approved formula based price cap control mechanisms. The prices below do not represent a binding capped price for an individual quoted service.

Table 1 Prices for quoted services (illustrative configuration only)

Quoted Service	2012–13 Price ¹ (\$ per service)
Rearrangement of network assets	4,348.14
Customer requested works to allow customer or contractor to work close	6,462.01
Non standard data and metering services (type 5-7 metering)	116.81
Emergency recoverable works and rectification of illegal connections	9,660.62
Large customer connections	363,798.05
Design specification/auditing and other subdivision activities	1,401.76
Unmetered services, including street lighting	1,884.67
After hours provision of any fee based service (excl re-energisation)	1,675.62
Supply abolishment - complex	464.90
Additional crew	123.20
Temporary connection - complex	47,560.58
Loss of asset	9,610.15
Other recoverable work ²	n/a

1. All indicative prices are exclusive of GST

2. There is no common configuration of the 'other recoverable work' service. The service is applied only in those circumstances where the service requested is not covered by any of the other service categories or would not otherwise have been requested for the efficient management of the network.



Appendix 2.4

CONFIDENTIAL

Price development for Alternative Control Services 2012-13

Appendix 2.5

CONFIDENTIAL

Additional Reporting Requirements for Alternative Control Services (ACS)



Appendix 3

Tariff Cost Allocation Process

Appendix 3

1 Tariff class cost allocation process

Energex's revenue cap (as determined by the AER) is based on a building block approach, which includes each of the following regulated cost components:

- Regulatory depreciation;
- Return on capital;
- Operating expenditure;
- Tax allowance;
- Capital contributions; and
- Revenue from shared assets.

The amount attributed to each component is provided in Table 4.1 of the Energex Pricing Proposal.

The purpose of the tariff cost allocation process is to allocate or assign the network costs to the tariff classes in the most efficient and cost reflective way. Section 1, of this Appendix sets out the process for the allocation of costs to tariff classes.

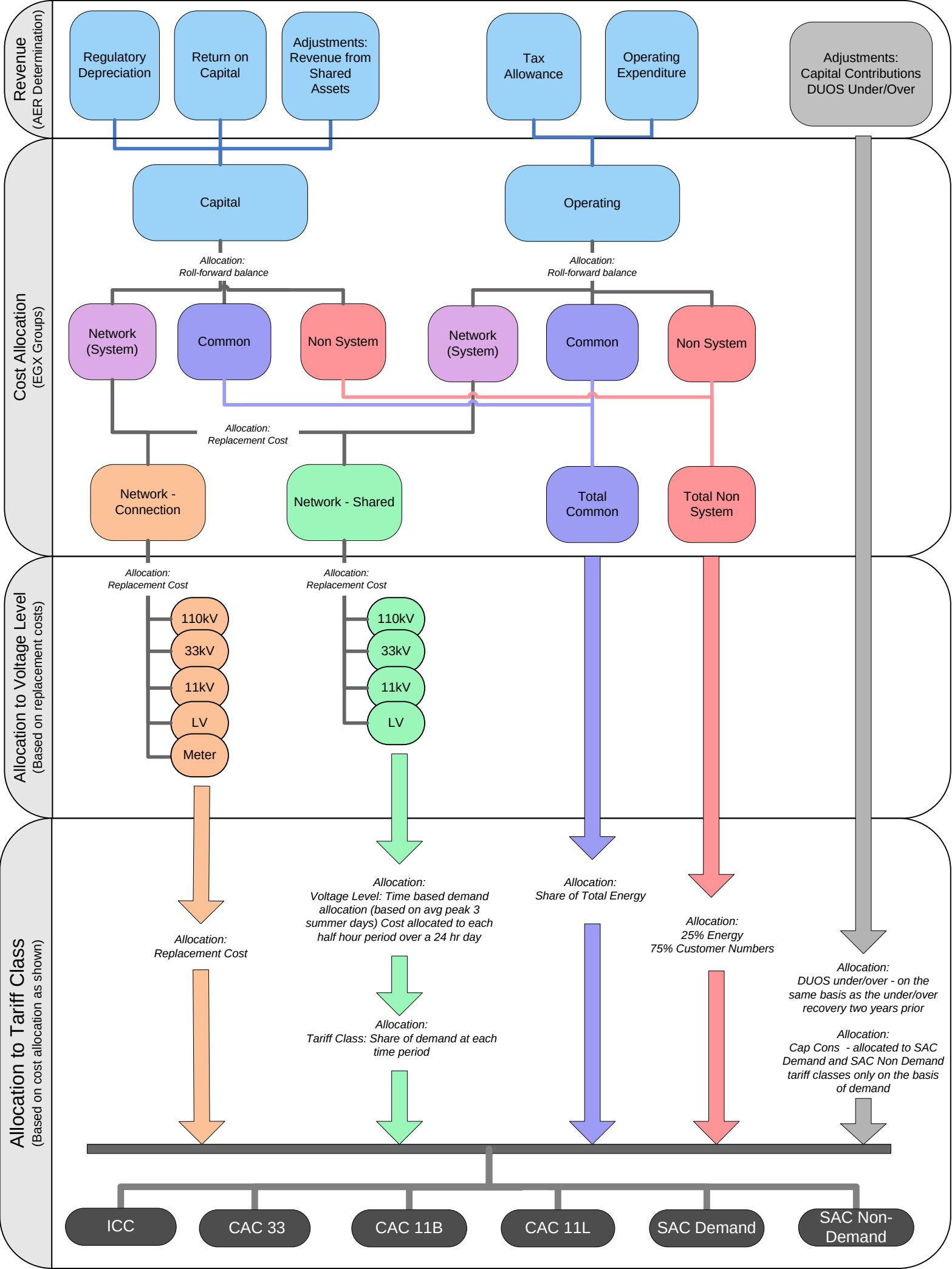
The major steps in the process are outlined in Figure 1, and described further below.

- Step 1 – Allocate AER building blocks to Energex DCOS cost groups
- Step 2 – Allocate Network (System) costs to voltage level
- Step 3 – Allocate costs to tariff classes



Figure 1 Cost Allocation to Tariff Classes

ENERGEX Tariff Cost Allocation Methodology – Tariff Class



Note on ICC

- **Note:** For ICCs the connection asset cost allocation includes the cost of their connection assets plus their proportion of the shared network based on a time based cost profile. This is discussed in section 1.2.1.

1.1 Step 1 Allocate AER building blocks to Energex DCOS Cost Groups

The regulated cost components specified by the AER are initially allocated into the Energex cost groups of

- Network (System);
- Common Services;
- Non System; and
- Adjustments

1.1.1 Network (System)

Network (system) costs are the directly attributable costs associated with the provision of network connection and distribution services that are attributable to a single customer or group of customers.

Network costs are allocated between connection assets and shared network assets based on the replacement cost of assets.

Network costs are further allocated to each of the following voltage cost groups based on the replacement costs of assets:

- 110/132kV;
- 33kV;
- 11kV;
- low voltage;and
- meters.

1.1.2 Common

Common Services costs are costs associated with those system assets that benefit the system as a whole and are not directly related to any single customer or group of customers. Assets included in this category are reactive plant, load control, control centres and communications.

1.1.3 Non-system

Non-system costs include corporate support (e.g. CEO, Finance, Human Resources, Legal), customer services, and occupancy costs which are not directly attributable to the operation and maintenance of the network but which are associated with network service delivery. These costs are treated consistently as a group as the cost drivers for this set of costs are consistent and it is impractical to manage a cost allocation stream for each of the specific components.



1.2 Step 2 Allocate Network (System) Costs to Voltage Level

1.2.1 Individually Calculated Customers (ICCs)

The cost allocation to each ICC is performed on an individual basis. Connection assets are assigned to ICCs based on information obtained from Energex network panel diagrams. Each ICC is then allocated a share of upstream shared network. This is based on the ratio of the customer's individual demand to the total demand of the respective supply (substation). The share of network costs is based on a time based profile with the demand for the substation and the customer allocated across daily half hour periods (48 periods) for the three peak summer days of the previous calendar year.

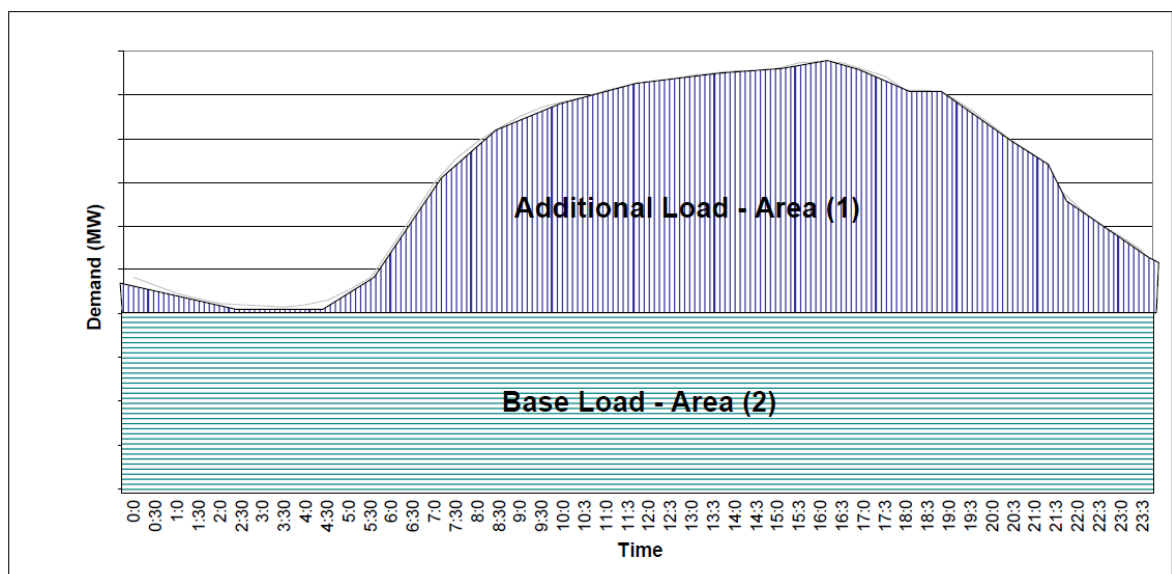
The ratio X in each period (Customer demand / Substation demand) is then applied to the replacement cost of system assets within the supply network to which the individual customer is connected. This is then aggregated to provide a total shared asset value for the customer.

1.2.2 Remaining allocation

Following the allocation of costs to ICCs, the remaining network costs are allocated to each of the voltage levels; 110kV, 33kV, 11kV and LV on a time basis.

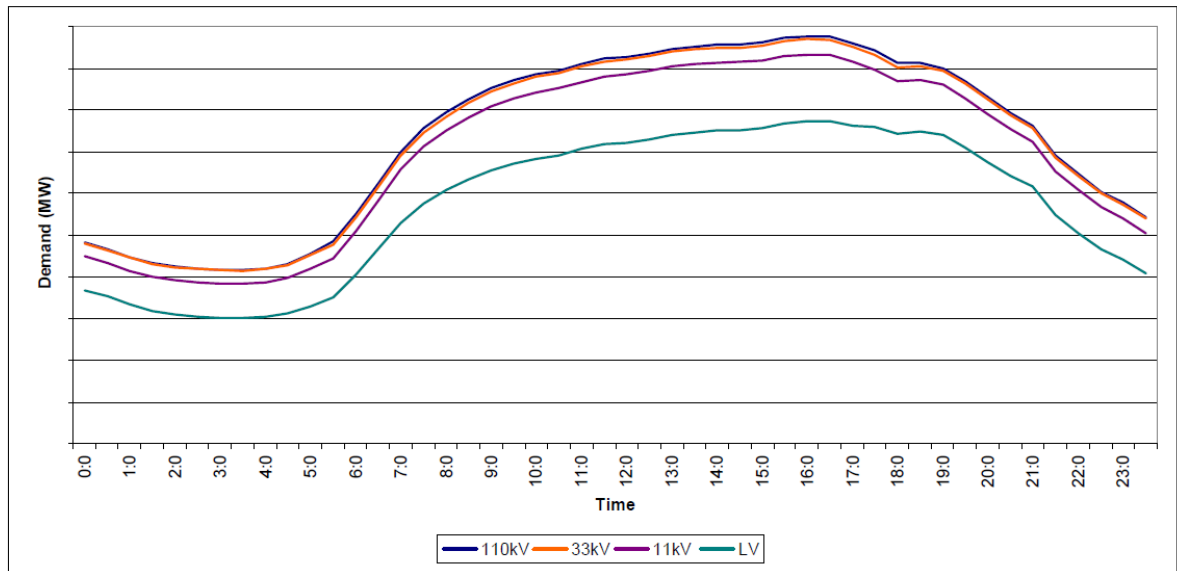
Network costs (capital and operating) are allocated to time periods based on the level of demand for that period. The basic principle is that the cost of providing the network (for system costs) should be allocated to the time periods it is being utilised. Accordingly there is a “base load” that uses a proportion of the network all the time (Area 2 on Figure 2). The cost of this base load of network can be recouped across all time periods. On the other hand, the extra network installed for other periods, including the peak, should be recouped across the time periods when this is used (Area 1 on Figure 2).

Figure 2 Time based cost allocation method



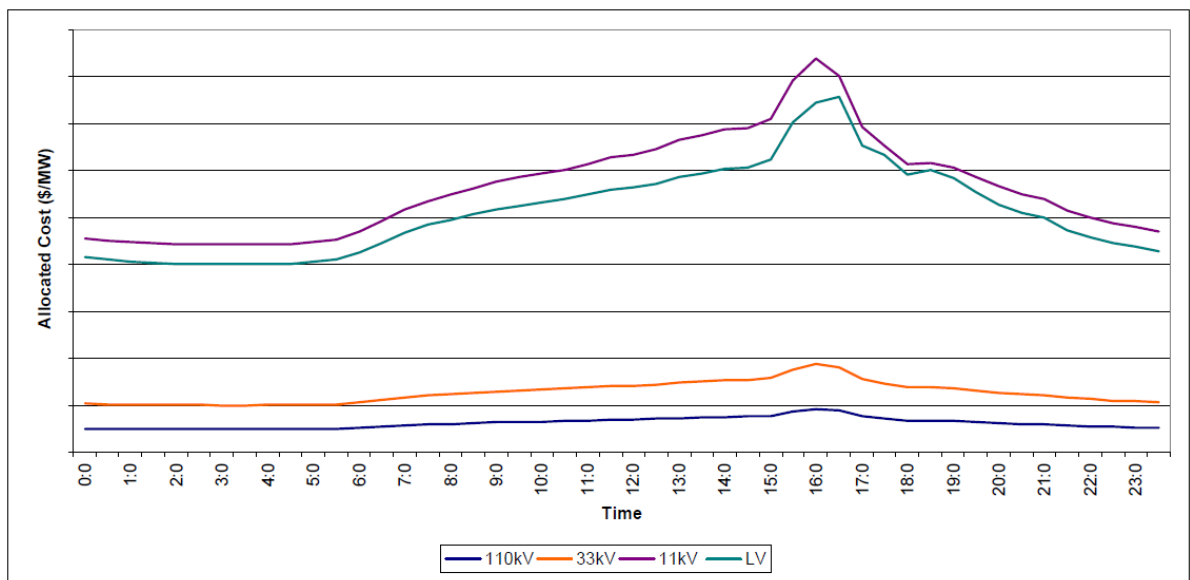
Meter data is obtained for the three peak summer days of the previous calendar year for all interval metered customers (110kV, 33kV, 11kV, LV Demand). From this meter data, an estimated peak daily load profile at each voltage level is obtained.

Figure 3 Demand profiles – Voltage Level



Costs are allocated across daily half hour periods (48 periods) based on an estimate of the peak daily load profile for each voltage level. More costs are allocated to periods with a higher demand on the network.

Figure 4 Daily cost allocation - Voltage Level \$/MW



1.3 Step 3 Allocate costs to Tariff Classes

There are 7 tariff classes (for standard control services) to which network costs are allocated:

- ICCs;
- CAC – 33kV;
- CAC – 11kV Bus;
- CAC – 11kV Line;
- EGs;
- SAC – Demand; and
- SAC – Non-demand.

1.3.1 Network

Connection Assets

Connection assets are allocated to each tariff class based on their share of the replacement cost for each voltage level. Contributed connection assets are not used in the allocation of capital costs as these assets have already been paid upfront by the customer.

Shared Network

With the exception of ICCs, the cost for each voltage level during each time period is allocated to each tariff class based on their share of demand at that time. This results in a daily cost allocation for each tariff class for each time period. The total across all periods is then aggregated as the cost allocated to each tariff class as shown in Figure 4.1 (Section 4 of Main Document).

A period demand (time of use) allocation has been used for the shared system costs as it reflects the fact that demand is the primary driver of shared network costs. By using a period demand it allocates costs of the network to the time periods when it is being utilised. This method reflects the impact that demand (in particular peak demand) has on increasing network investment.

Figure 5 Demand Profiles – Tariff Class

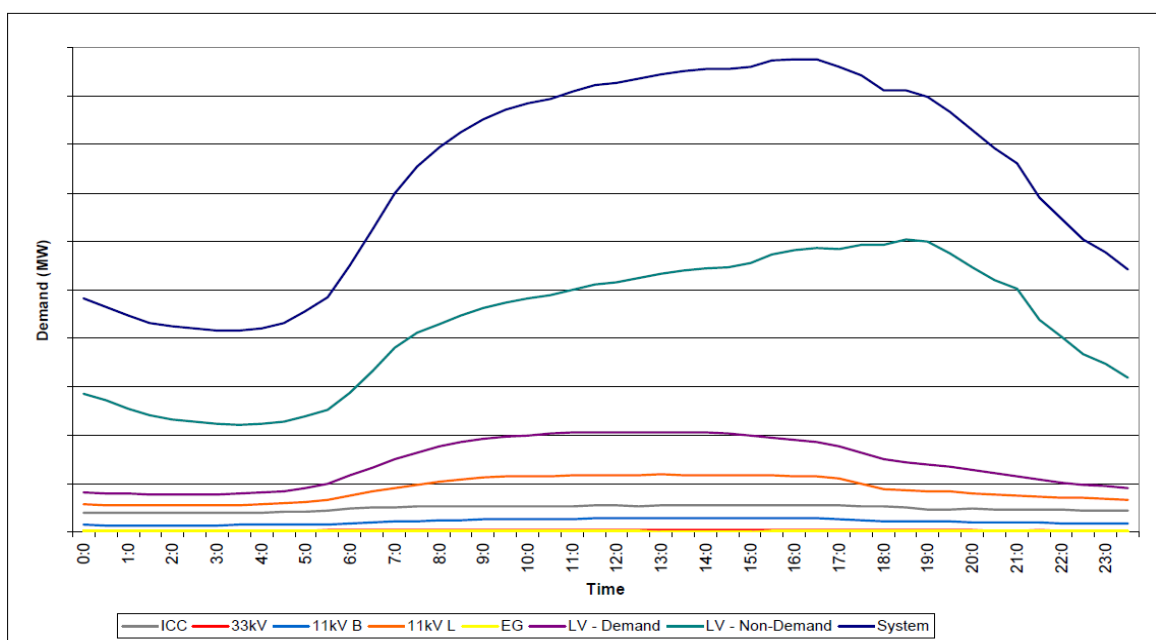
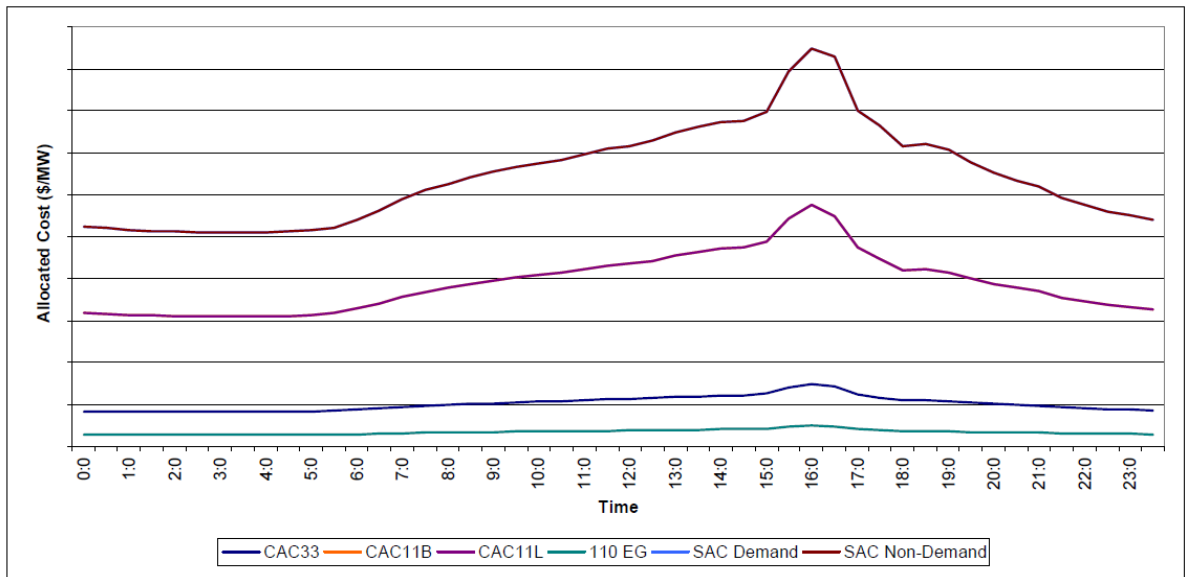


Figure 6 Daily cost allocation – Tariff Class \$/MW



1.3.2 Common Services

Common services costs are allocated to each tariff class using total energy as the allocation method.

Anytime energy is used in cost allocation for costs which are related to the size of the customer but not specifically the demand the customer places on the network.

1.3.3 Non-system

Non-system costs are allocated to each tariff class using a hybrid allocation. Customer numbers (75%) and total energy (25%) is used. These weightings reflect that the number of customers is the primary driver of non-system costs.

Customer numbers and energy are used for the cost allocation method as these costs are associated with the number of connected customers and their expectations/service requirements. Energen has a number of costs that are customer number based. A significant proportion of the overhead costs of the business are driven by the number of staff and systems required to serve the customer base.



2 Tariff (charging parameter) cost allocation process

Following the cost allocation to tariff classes, costs must be allocated to tariffs and ultimately to charging parameters (tariff elements), which may include any combination of the following:

- Fixed charges;
- Capacity charges;
- Demand charges; and
- Energy charges.

The purpose of the tariff (charging parameter) cost allocation process is to allocate or assign the costs to each parameter in the most efficient and cost reflective way. Section 2 of this Appendix sets out the process utilised by Energex for the allocation of costs to charging parameters.

The major steps in the process are outlined in Figures 7,8,9.

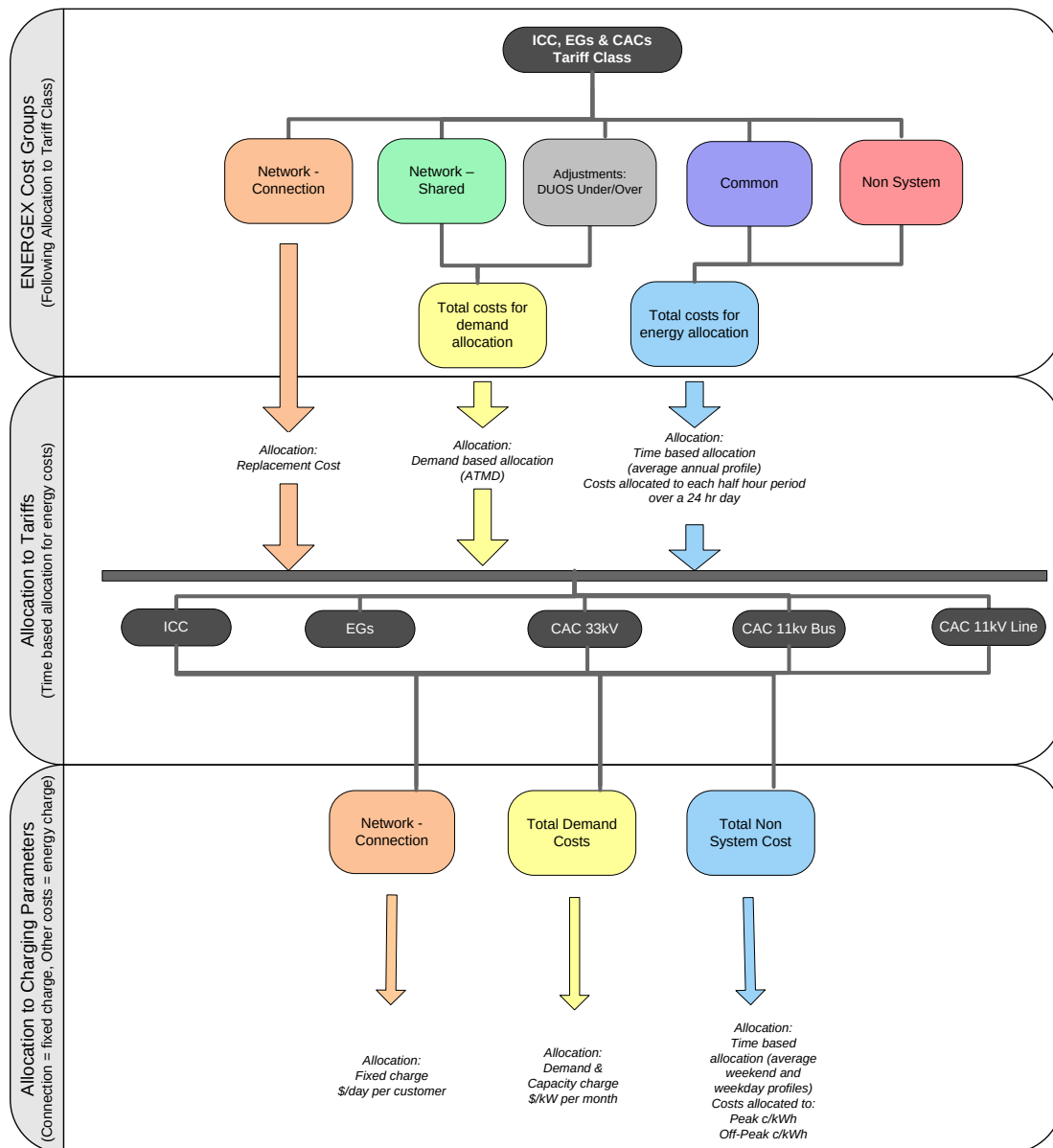
- Step 1 – Allocate tariff class costs to individual tariffs
- Step 2 – Allocate tariff costs to charging parameters



2.1 Individually Calculated Customers (ICCs), Embedded Generators (EGs) and Connection Assets Customers (CACs)

Figure 7 Tariff Allocation Method – ICCs, EGs & CACs

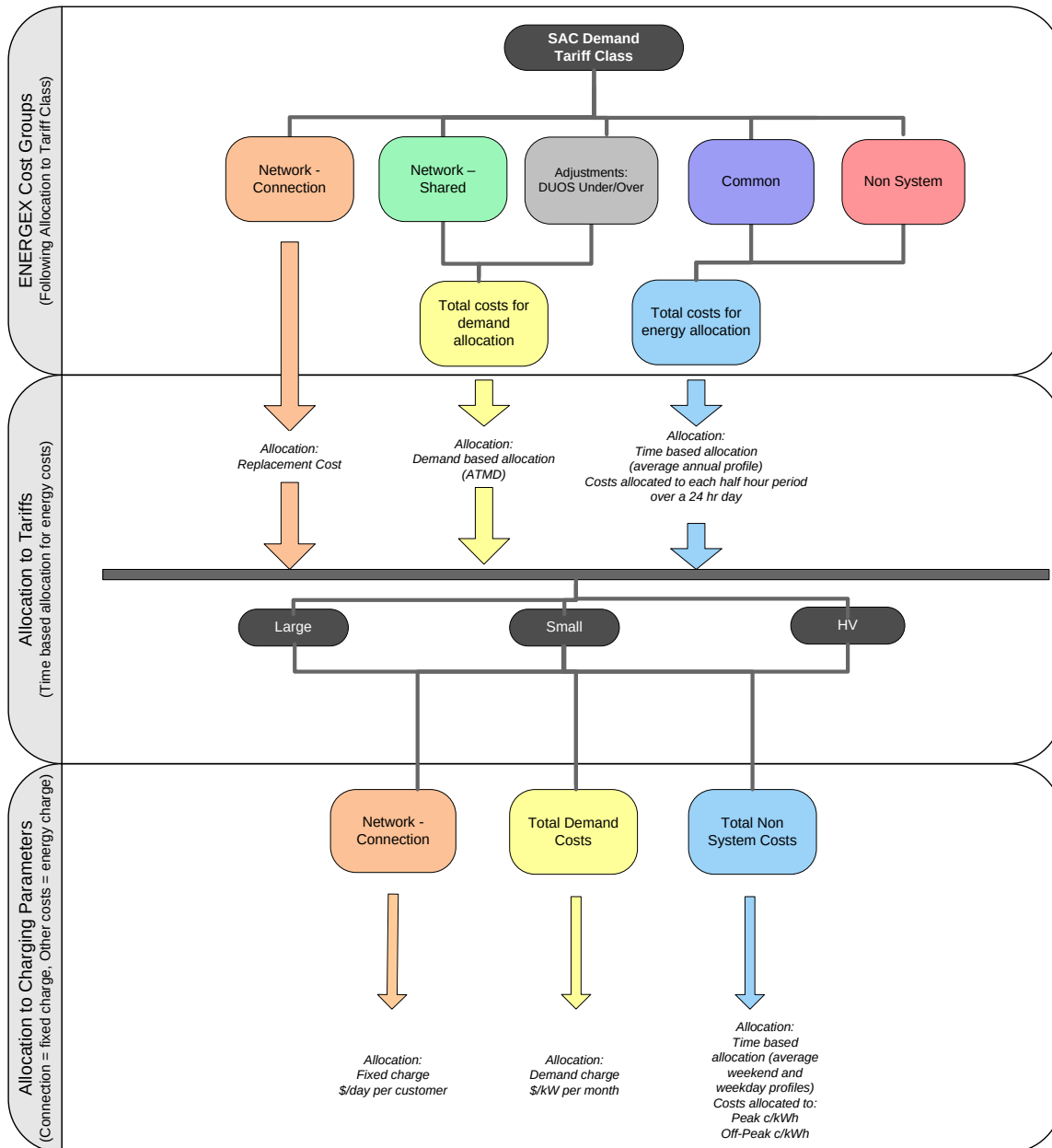
ENERGEX Tariff Cost Allocation Methodology – ICC, EGs & CACs Tariffs



2.2 Standard Asset Customers – Demand

Figure 8 Tariff Allocation Method – SAC Demand

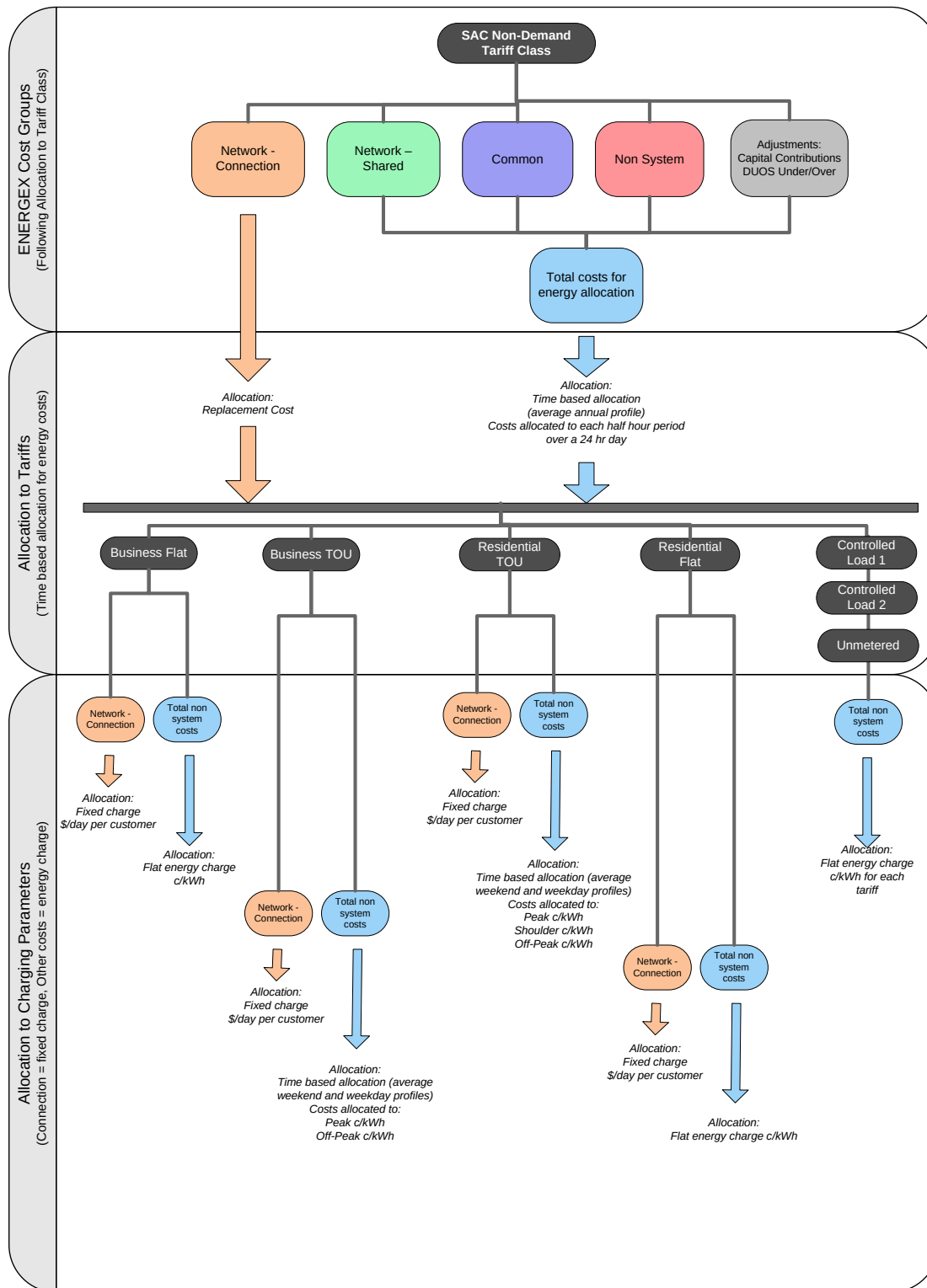
ENERGEX Tariff Cost Allocation Methodology –SAC Demand Tariffs



2.3 Standard Asset Customers – Non-Demand

Figure 9 Tariff Allocation Method – SAC Non-Demand

ENERGEX Tariff Cost Allocation Methodology – SAC Non-Demand Tariffs



3 Variances from allocated cost based tariffs

Energex develops and applies network tariffs based on the distributed cost model. For each tariff class, the costs outlined above are recovered through a combination of fixed charges, demand charges and/or volume charges. The network pricing methodology applied to each of those groups has precluded any possible bypass challenge on the basis that the network tariff is efficient and an alternative electricity service can not be sourced at a lower economic value. Providing it is consistent with the *Rules*, Energex may negotiate a tariff other than the tariff calculated using the cost allocated methodology where it can be demonstrated that the cost-based network tariff is not efficient, as an economic bypass opportunity exists or an alternative electricity service could be utilised.

